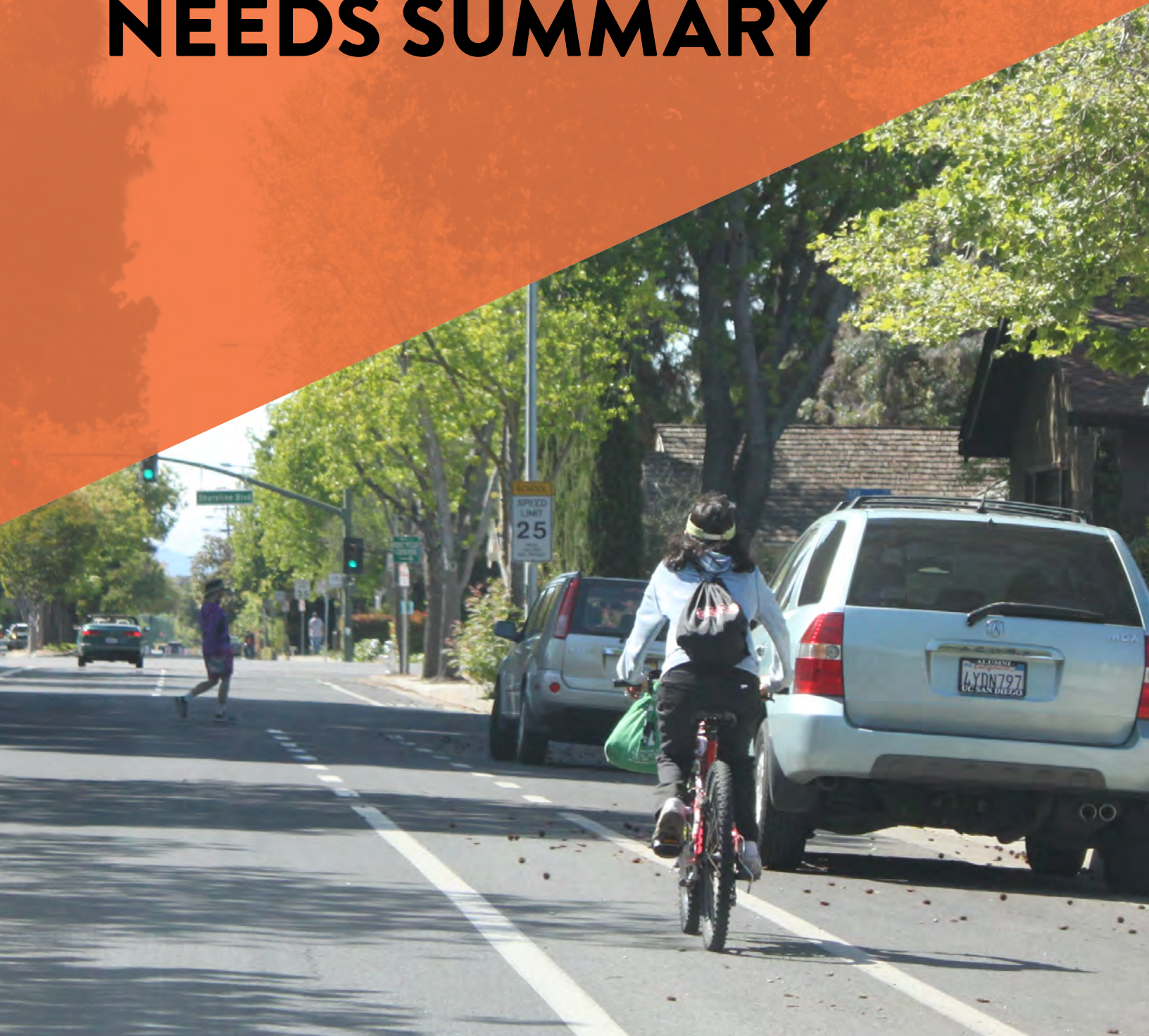


APPENDIX E: EXISTING CONDITIONS AND NEEDS SUMMARY



Appendix E

Existing Conditions and Needs Summary

October 2023
Revised July 2026



City of
Mountain View



Note

This report was written in 2023 and a draft was presented to the Mountain View Bicycle and Pedestrian Advisory Committee in [October 2023](#). The report should be viewed as a snapshot in time and may not be reflective of current conditions.

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INTRODUCTION

Mountain View's Active Transportation Plan

Mountain View envisions an active transportation network that is safe and comfortable for all users regardless of age, ability, income, race or language. The Mountain View Active Transportation Plan (MVATP) aims to identify citywide policies and projects that will improve walking (including wheelchairs or disability-related devices), biking, rolling (including e-scooters and other micromobility devices), and green streets in Mountain View.

The ATP is guided by a vision statement and five guiding principles:

Vision Statement:

Mountain View leads the region with an Active Transportation Plan that defines a series of connected, low-stress, and green corridors that invite active transportation. The corridors include shade, wildlife habitat, and public open spaces. The plan enables the City to intentionally support a culture of walking and biking, enhance biodiversity, and reduce climate change impacts.

Guiding Principles:

- Mobility & Connectivity
- Safety & Comfort
- Access & Equity
- Sustainability & Biodiversity
- Innovation & Action-Orientation



About this Report

The Existing Conditions and Needs Summary describes key aspects of existing infrastructure, demographics, and other community characteristics that influence walking, biking, and green streets in Mountain View. In addition to summarizing quantitative data, this report includes qualitative data in the form of public input and comments. Understanding existing conditions allows the City to identify what works well and areas for improvement. The data collected for this report was used to identify and prioritize recommendations for the Active Transportation Plan.

The report begins with an overview of community demographics and mode share in Mountain View and then uses four of the MVATP's guiding principles to understand current conditions for active transportation:

- Mobility & Connectivity
- Safety & Comfort
- Access & Equity
- Sustainability & Biodiversity

Data Sources

Data sources used in this report are summarized below. Data used in this report reflects what was available in 2022-2023 when the report was drafted.

City Databases

Data from the City of [Mountain View's open data portal](#) provided information on physical infrastructure including street centerlines, sidewalks, crosswalks, bicycle facilities, posted speed limits, land uses, tree canopy, stormwater, current and planned projects, shuttle services, and public safety records.

Census Data

Community demographics were collected from the [U.S. Census American Community Survey \(ACS\)](#) 5-year estimates.

State Crash Database

Historical crash data was downloaded from the [Transportation Injury Mapping System \(TIMS\)](#), which is a geodatabase of police-reported crashes in California, from the California Statewide Integrated Traffic Records System..

Transit Service Data

The Santa Clara Valley Transportation Authority (VTA), Mountain View Transportation Management Association (MTMA), and Caltrain provided data on transit stops, routes and ridership.

Community Input

Community input is an important data source to understand the existing conditions, which are not only about what is physically on the ground or in a policy but also how it is experienced by those who travel. The City deployed a multi-pronged approach to engagement during the existing conditions phase of the MVATP, recognizing that people vary in their comfort or preference for providing input. A mix of digital, print and in-person opportunities at various times and with interpretation helped to make engagement open and

inclusive. Additional multilingual engagement was conducted to improve response rates from under-represented groups including Spanish-speaking communities.

Engagement efforts are summarized below.

ATP Survey

An online mapping survey was developed in the spring of 2023 using a survey platform called [Maptionnaire](#). The survey asked participants a series of questions about their experiences walking and rolling in Mountain View—their favorite places and routes, their challenging spots and routes and what and where they would like to see improvements. Participants were asked to digitally record their answers on maps of the city of Mountain View and to share any additional comments. Sociodemographic information was optional.

The survey, available in English, Spanish, Chinese and Russian, was open for public participation from May 15 – July 31, 2023. It was promoted via postcards, flyers and lawn signs distributed around Mountain View, digital social media promotions and email blasts. (Figure 1) Six hundred fifty-five people participated in the survey.



Figure 1: The ATP Survey was promoted in four languages using print and digital communications

In-person meetings/presentations

Pop-up and in-person meetings were held to connect with communities that often do not participate as frequently in engagement activities in Mountain View. The meetings gathered input on participant experiences walking, biking and rolling in Mountain View. These events are listed below:

- Thursday June 15, 2023, 4:30–7:30 p.m.: Community Services Agency (CSA) (with Spanish interpretation)
- Monday, July 17, 2023, 8:30–9:30 a.m.: Mountain View Dayworker Center (Spanish interpretation)
- Tuesday, August 22, 2023, 11 a.m.–1 p.m.: Mountain View Senior Center
- Wednesday, July 26, 2023, 2:00–3:30 p.m.: The View Teen Center.

Bike Tours

Two bike tours were held to gather community input on the bicycling experience in Mountain View. Approximately 40 adults and three children participated in the three-hour, 11-mile bike tours, which focused on areas identified as important bicycle connections or challenging areas in Access MV, Ask MV, and survey comments. The events were held:

- Saturday, May 13, 2023, 10:00 a.m.–noon: South of Caltrain
- Sunday, September 23, 2023, 10:00 a.m.–noon: North of Caltrain

Walk Tours

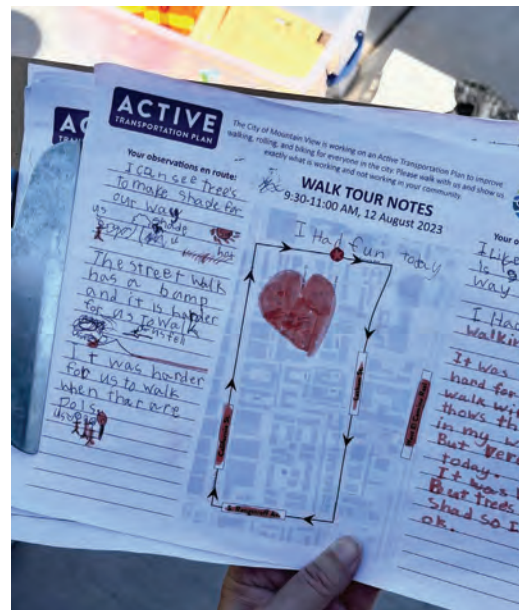
Three walk tours were held to gather community input on the walking experience in Mountain View. A park, a school and an area known to have more seniors were intentionally selected to understand their experiences. Events were held:

- Friday, August 11, 2023, 6:00–7:30 p.m.: Rengstorff Avenue, Latham Street and El Camino Real (27 participants)
- Saturday, August 12, 2023, 9:30–11:00 a.m.: Escuela Avenue, California Street and Latham Street (10 participants)
- August 25, 2023, 5:30–7:00 p.m.: Sylvan Avenue, Glenborough Drive and Moorpark (22 participants)

AskMV

The City of Mountain View maintains an online comment platform called [AskMV](#), where individuals can share on a variety of topics. City staff monitor the platform for comments related to walking and bicycling experiences, recommended improvements and

Figure 2 Worksheet from Walk Tour



The walk tours were organized to solicit a wide range of views from the community, such as this young student's perspective

transportation safety. Approximately three years of feedback was gleaned to identify community concerns.

BPAC

Mountain View [Bicycle and Pedestrian Advisory Committee](#) (BPAC) members are appointed by the City Council to represent the interests of people who walk and bike in the city. BPAC reviewed and provided input on ATP-related topics at the following meetings:

- [Tuesday, November 30, 2021](#) – Review and comment on proposed task list for MVATP
- [Wednesday, February 22, 2023](#) – Receive report on scope of work for MVATP
- [Wednesday, October 25, 2023](#) – Receive draft existing conditions report
- [Wednesday, April 24, 2024](#) – Review and provide feedback on MVATP scoring criteria
- [Wednesday, March 11, 2025](#) – Recommend holistic network maps and scoring criteria to City Council
- [Wednesday, April 29, 2026](#) – Review and comment on the Public Draft MVATP

Active Transportation Plan Advisory Committee

The Active Transportation Plan Advisory Committee (ATPAC) was formed to have an external advisory group that can provide input to the ATP as it is developed and act as a liaison for their networks. Members represent non-profit organizations and public agencies that work in active transportation, green streets or serve the needs of those that walk and roll, such as but not limited to, the Community Services Agency, Santa Clara Public Health Department, VTA, the Day Worker Center, MV Mobile Home Alliance, Youth Advisory Committee, Mountain View Los Altos High School District, Silicon Valley Bicycle Coalition, and Mountain View Chamber of Commerce. ATPAC members helped organize and participate in many of the engagement activities.

1. DEMOGRAPHICS AND MODE SHARE

Demographics

The City of Mountain View is a Santa Clara County community that experiences the opportunities and challenges of a strong Bay Area economy. Based on the 2020 U.S. Decennial Census, the city's population is approximately 82,000, an increase of over 10% since 2010. Additionally, the influx of professionals to public and private employers increases the city population by over 100,000 during most weekdays. Approximately 55 percent of the population identifies as male and 45 percent as female. The city is relatively young, with almost 67 percent of its population under 44 (median age 35.3 years) and approximately 22% under the age of 18 (U.S. Census 2021 American Community Survey [ACS] 5-year estimates)—a demographic for which a robust and safe active transportation network provides an opportunity for independent mobility (Table 1).

Table 1 Mountain View Age Demographics

Age	Number	Percent
Under 5	5,238	6.4
5 to 14 years	9,113	11.1
15 to 19 years	3,720	4.5
20 to 24 years	4,343	5.3
25 to 44 years	32,655	39.6
45 to 64 years	18,192	22.1
65 to 79 years	6,624	8.0
80 and over	2,524	3.1
Total	82,409	100.0

ACS 2021 5-year estimates

Mountain View's overall population density is low to medium, with some areas in the central west portion of the city between Central Expressway/Caltrain and El Camino Real being of higher density (Figure 3). Those neighborhoods also tend to have higher youth density (Figure 4).

Figure 3 Population Density

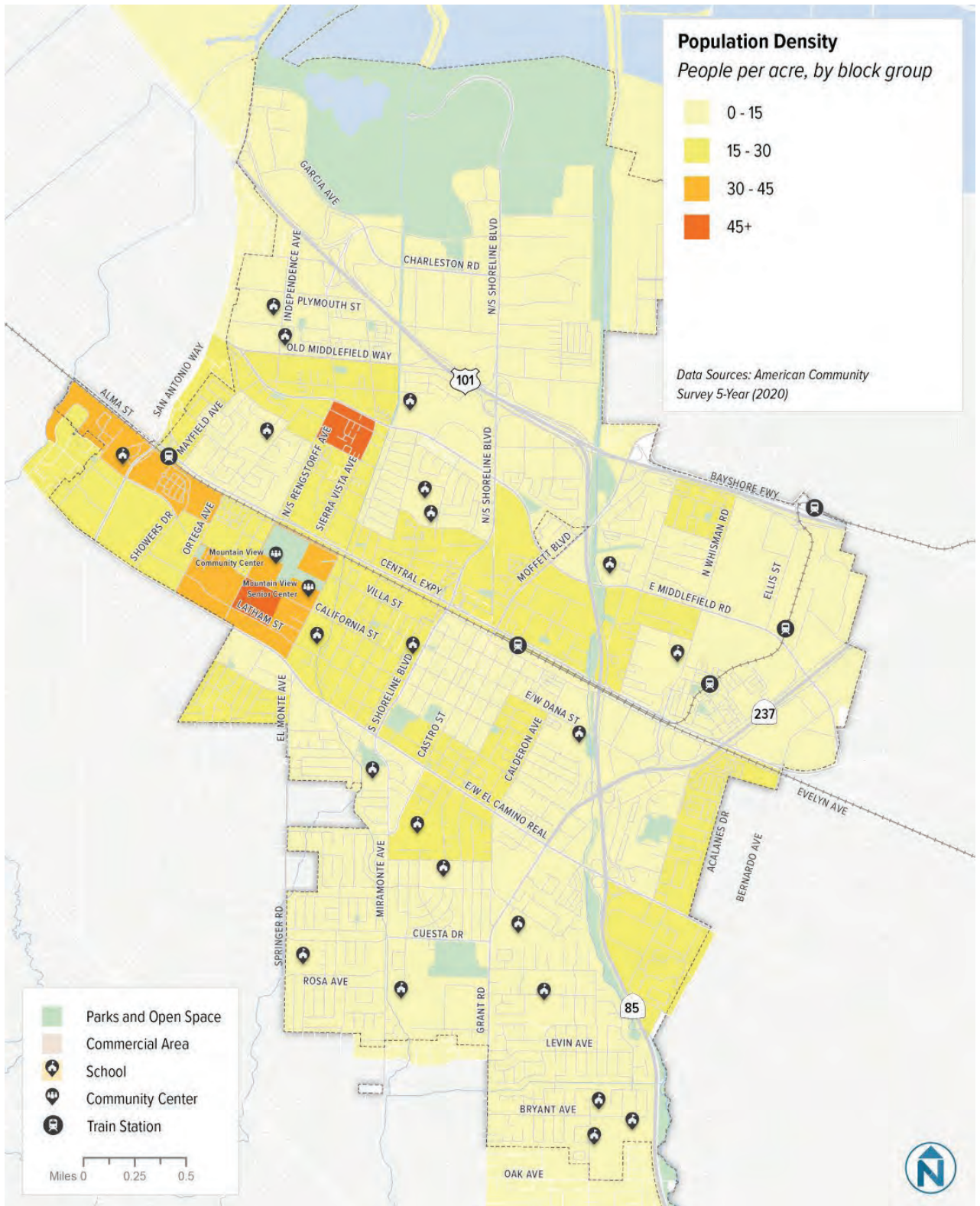
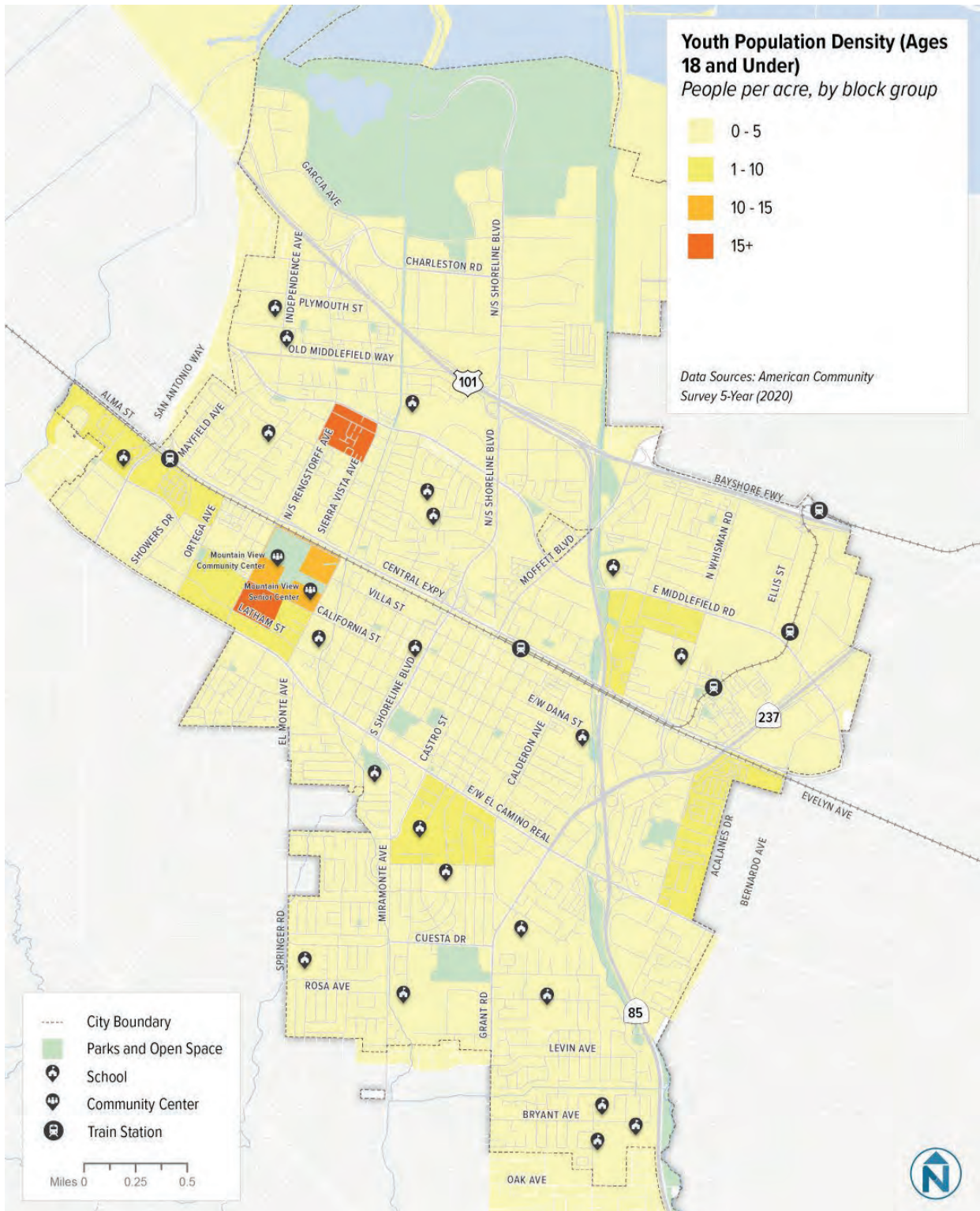


Figure 4 Youth Population Density (Ages 18 and Under)



According to the 2021 ACS 5-year estimates (Table 2), approximately 48 percent of the population of Mountain View identifies as white, 33 percent identify as Asian (with most identifying as Chinese), and 18 percent identify as Hispanic or Latino.

Table 2 Race and Ethnicity in Mountain View (2021)

Race and Ethnicity in Mountain View	% of Residents
White alone	48.4%
Asian alone	33.3%
Black or African American alone	2.4%
Native American, Native Alaskan alone	0.4%
Native Hawaiian, Pacific Islander alone	0.0%
Some other race alone	5.8%
Two or more races	9.6%
Hispanic or Latino Population in Mountain View	% of Residents
Not Hispanic or Latino	82.0%
Hispanic or Latino:	18.0%

ACS 2021 5-year estimate

Approximately 5.5 percent of the community is below poverty level. The median household income in Mountain View is \$158,104, while the mean is \$213,751 (Table 3). As of August 2023, the median home price in Mountain View is \$1.79 million.¹

Table 3 Household Income Distribution in Mountain View

Income range	% of Residents
Less than \$50,000	16.8%
\$50,000 to \$99,999	15.8%
\$100,000 to \$149,999	15.2%
<i>Median household income: \$158,104</i>	
\$150,000 to \$199,999	13.0%
\$200,000 or more	39.2%

ACS 2021 5-year estimate

¹ <https://www.redfin.com/city/12739/CA/Mountain-View/housing-market>, retrieved October 2023

Mode Share and Vehicle Access

Approximately 5.6% of workers 16 years and over in Mountain View bike to work (compared with 0.8% statewide and 0.5% nationally), with males riding at a higher rate than females (Table 4). Approximately 3.2% of workers reported walking to work (compared with 2.4% statewide and 2.5% nationally), with females walking at a slightly higher rate than males. More than half of Mountain View residents commute to work by driving alone, while over 30% work from home. Commute mode share in Mountain View and peer cities is discussed further in Part 4 and summarized in Table 7.

Figure 5 and Figure 6 show the geographic distribution of walking and biking commute behavior in Mountain View. While the Journey to Work data set of the U.S. Census only provides a snapshot of rates of walking and bicycling within the city by one trip purpose, it indicates that even with that limited trip focus, travel by active transportation modes are utilized in the city.

Table 4 Commute Mode Share in Mountain View (2021)

Commute Mode	Mountain View	California	United States
Drove alone	53.2%	70.1%	73.2%
Carpooled	4.5%	9.6%	8.6%
Public transportation	2.1%	4.1%	4.2%
Walked	3.2%	2.4%	2.5%
Bicycle	5.6%	0.8%	0.5%
Taxicab, motorcycle, or other means	1.2%	1.6%	1.4%
Worked from home	30.2%	11.4%	9.7%

Source: ACS 2021 5 Year Estimate

Almost 5% of households in Mountain View do not have a motor vehicle available (2021 ACS 5-year estimates). In looking at this data spatially, some block groups in Mountain View report up to 24% of households without access to a vehicle (Figure 7). Some of these block groups mirror those with higher percentages of population and youth density and bicycling to work.

Figure 5 Mode Share – Percent of Commuters Who Walk

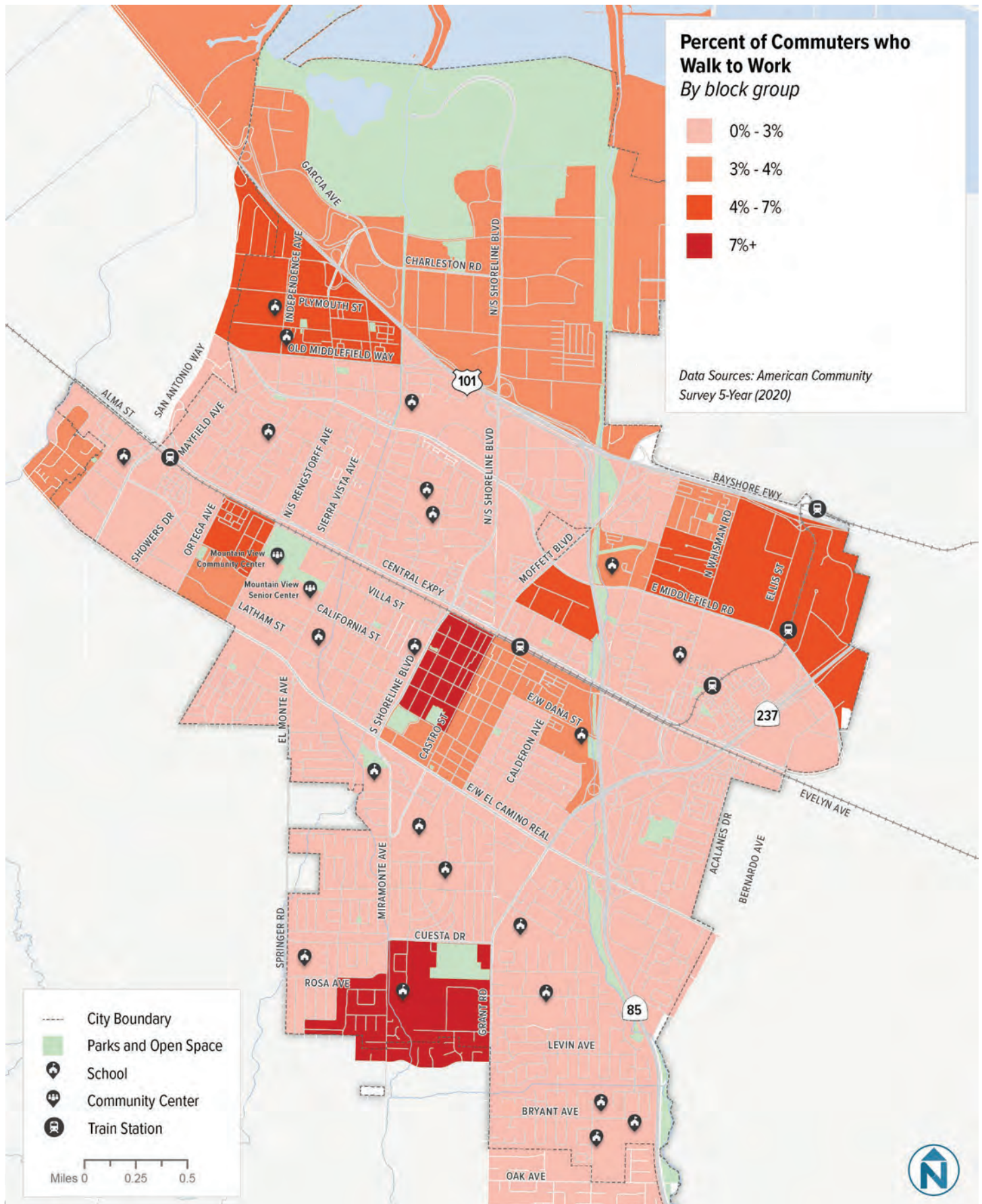


Figure 6 Mode Share – Percent of Commuters Who Bike

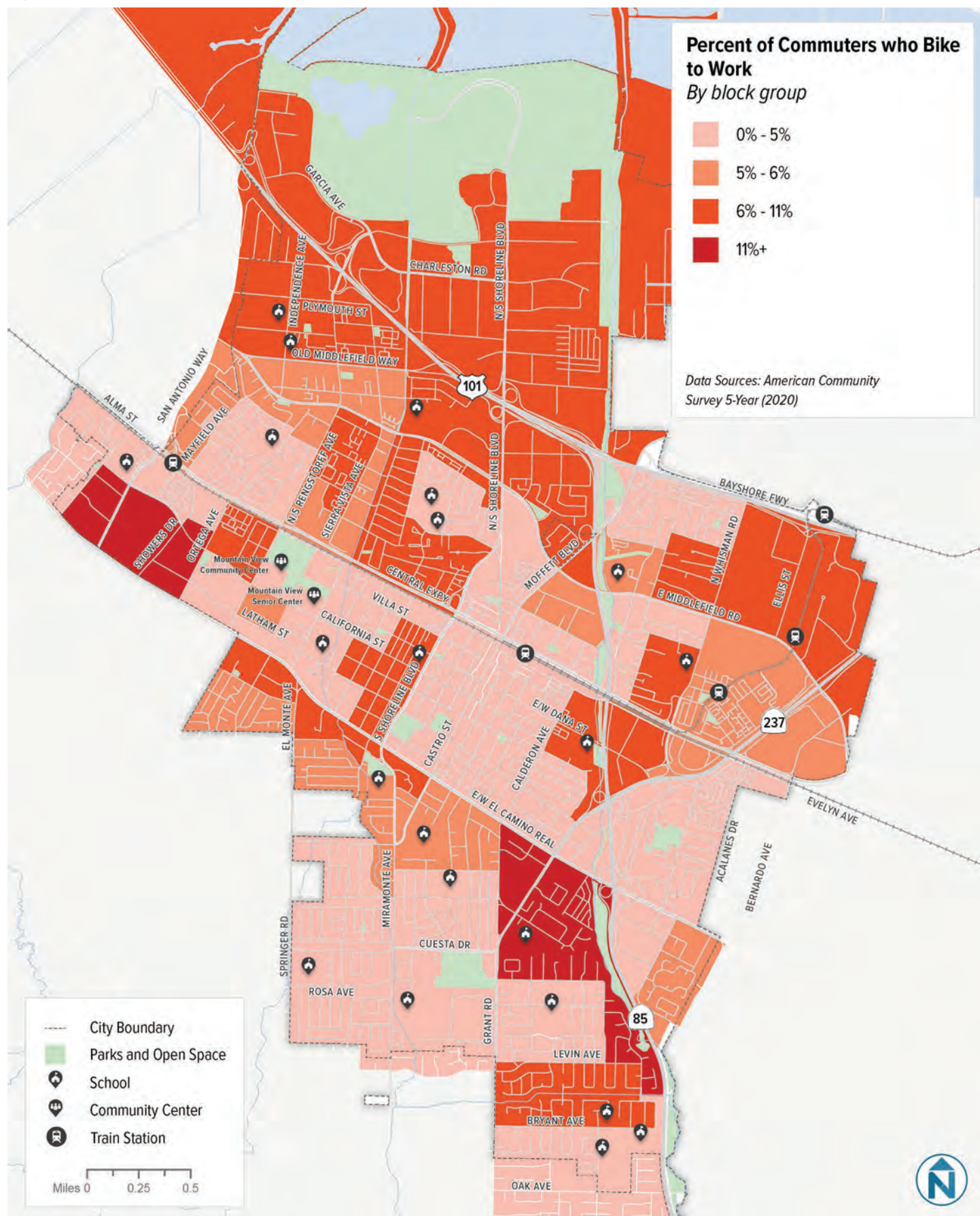
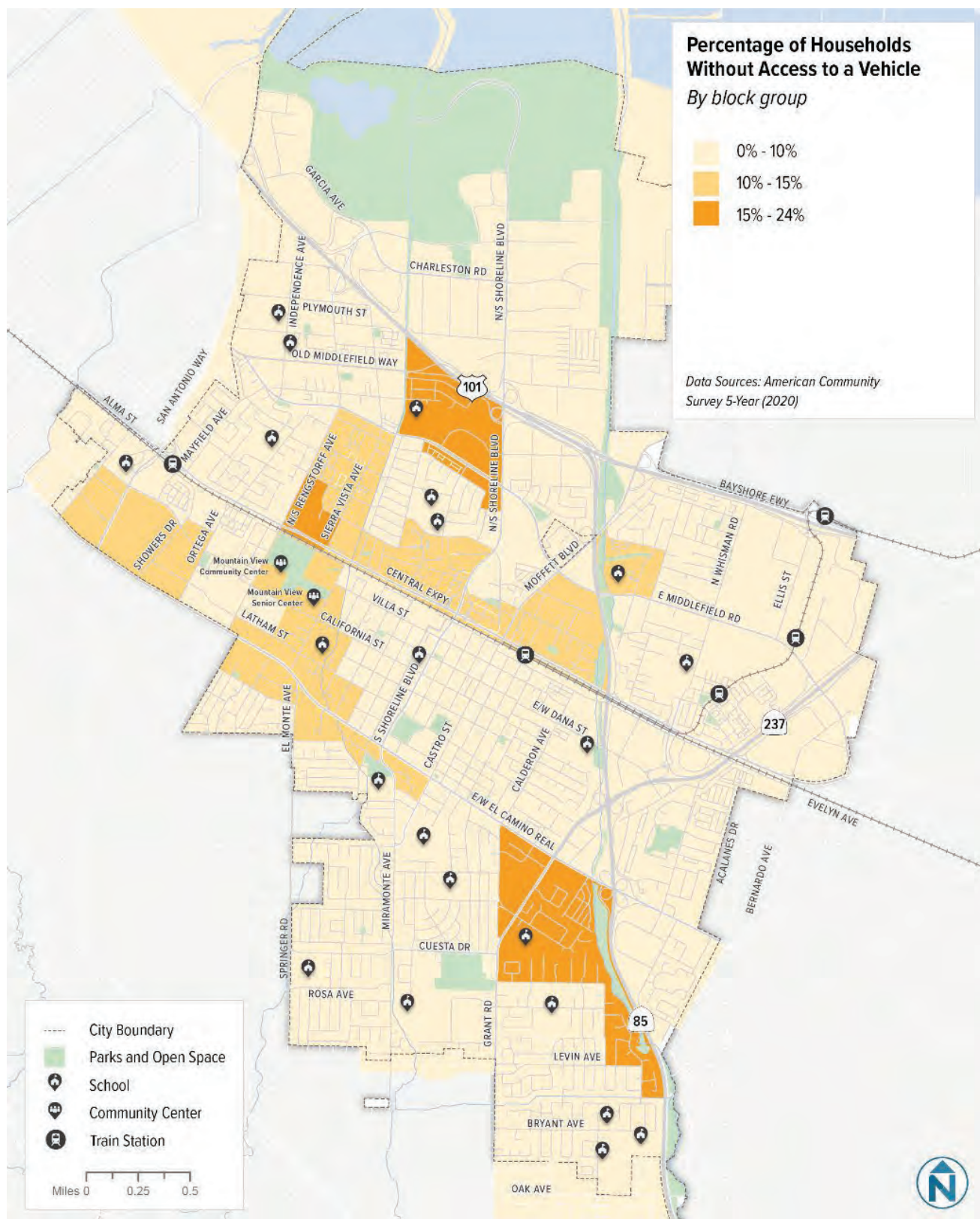


Figure 7 Household Vehicle Access



2. MOBILITY AND CONNECTIVITY

In Mountain View, pedestrian and bicycle networks are essential components of the transportation system. Evaluating the coverage and continuity of these networks can help the City identify where improvements are needed to better support the daily needs of everyone who lives, works, goes to school, and visits the City.

Pedestrian Network Coverage and Gaps

Mountain View has a well-connected pedestrian network that supports walking and rolling throughout most of the city. Figure 9 shows the extent of the existing sidewalk and pedestrian network in Mountain View today, including sidewalks, trails, and pedestrian paseos, and highlights places where there are gaps, or missing segments in the sidewalk network, along with areas where the sidewalk is less than four feet in width.

Sidewalks

The city's sidewalk network is 95% complete. However, there are a few neighborhoods with missing and narrow sidewalks, such as Moorpark Way and along Wright Avenue, or in the southern portion of the city, such as west of Miramonte between Marilyn and Barbara, and southeast Mountain View between Grant Road and SR-85.

Many areas with missing sidewalks were annexed from the County and did not include sidewalks at the time of annexation. The Policy on Unimproved Streets adopted by City Council in 1993 affirms the direction to not construct sidewalks in residential areas unless homeowners initiate an assessment district to fund the sidewalks. The policy references resident desires to maintain the rural character of these streets by not improving streets to current standards.

Mountain View's mobile home parks (Santiago Villa, Sahara Mobile Village, New Frontier, Moorpark Mobile Homes, Moffett Mobile Home Park and Sunset Village) are privately owned and shown as areas of "no data" on Figure 8. Mobile home parks are typically constructed with private streets which do not include a separate sidewalk from the vehicle drive aisle. While the City has limited control over internal street standards for existing private streets, there can be opportunities to better connect the surrounding City sidewalk network to the mobile home parks. Residents in these neighborhoods, where access to vehicles is lower (Figure 7), may rely more on walking and other non-driving transportation modes to meet daily travel needs.

Survey respondents and community members noted missing or incomplete sidewalks on one or both sides of Martens Avenue, Sleeper Avenue, Franklin Avenue, Eunice Avenue, and Carmelita Drive. These streets connect pedestrians from residential streets to nearby parks

and schools, such as Sleeper Park, Cuesta Park, Cooper Park, Stevens Creek Trail and Amy Imai Elementary School.

Survey participants also noted issues with sidewalks on Moffett Boulevard near the Stevens Creek Trail access point, such as sidewalk gaps, the condition of sidewalks, and the desire for an improved crossing to access the trail.

Paseos

Mountain View's pedestrian network includes more than just sidewalks and crosswalks—a host of paseos and trails also support connectivity and mobility throughout the City for people who walk (Figure 9). A paseo is a short connection, often utilizing a public access easement that allows for increased pedestrian and cycling connections. Paseos most commonly occur in the middle of very large blocks (Figure 8).

Paseos are a valuable tool to provide public permeability for pedestrians and cyclists through private areas. These are often built as part of multi-family residences in Mountain View. They have the added benefit of providing “eyes on the street,” which increases safety and provides a sense of community.

Participants in the Rengstorff Park Walk Tour noted how the paseo connecting Latham to El Camino between Rengstorff and Ortega improved access to destinations on El Camino Real.

Survey respondents suggested more pathways, or paseos, through apartment and office complexes near schools and on routes to transit, such as the Caltrain stations, to increase active transportation accessibility.

Figure 8 Examples of Paseos and Car-Free Connections in Mountain View

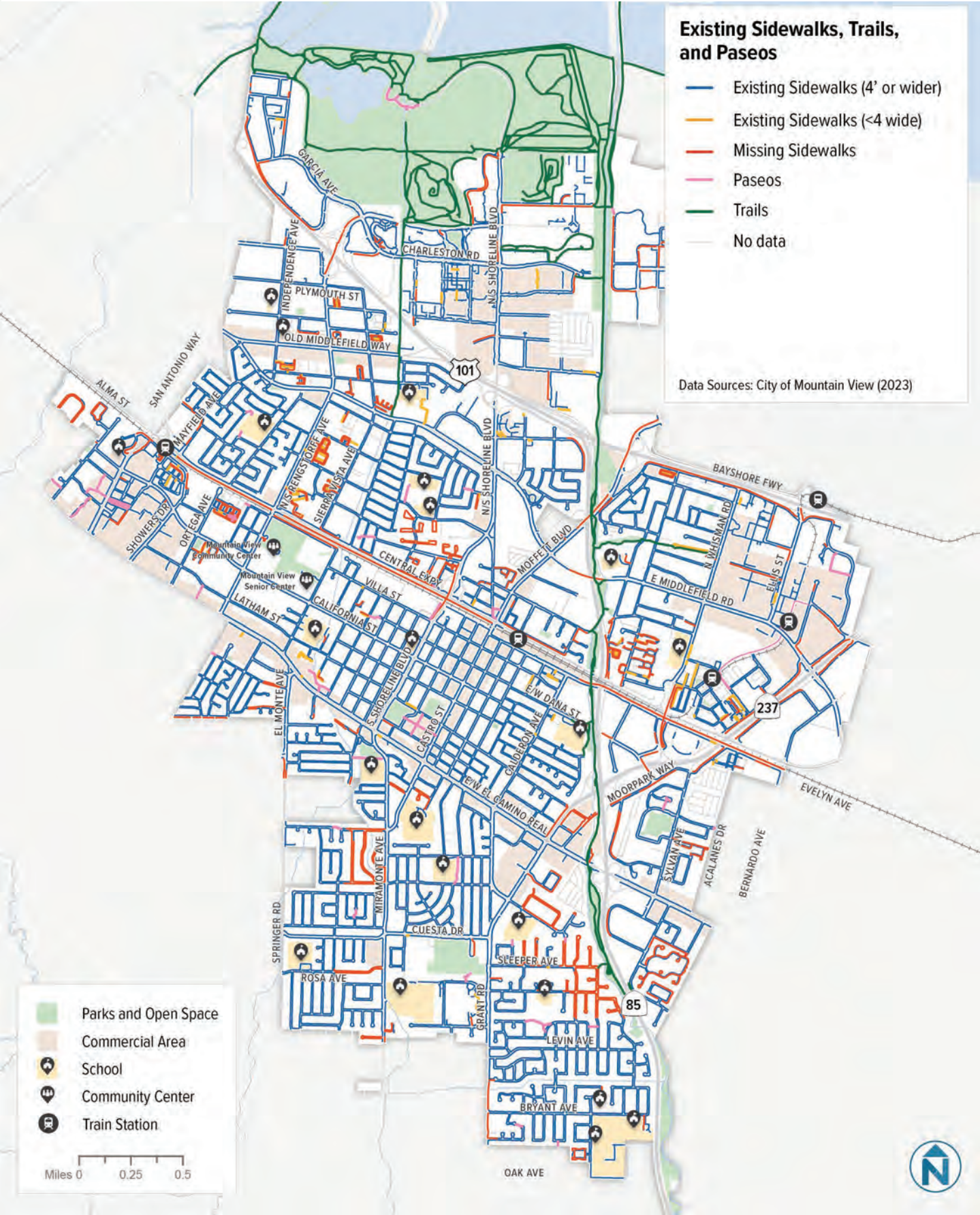


Paseos are a useful tool to provide public access through private areas.



This separated pedestrian and bike connection at the southern end of Sterling Street provides a car-free connection to Central Expressway at Moffett Blvd.

Figure 9 Pedestrian Network in Mountain View



Sidewalk Accessibility

As noted earlier, the city's sidewalk network is 96% complete, and many newer sidewalks provide comfortable pedestrian conditions. In some locations, however, there are obstructions (such as utility poles or signs) or poor physical conditions (such as uplift, missing curb cuts, narrow sidewalks). Some of these conditions are associated with the age of the sidewalks and standards in place at the time of construction.

The City of Mountain View is actively developing an ADA Transition Plan which will seek to identify non-compliant curb ramps, sidewalks, and other infrastructure in the right-of-way and develop a work plan to bring those locations into compliance.

Figure 10 High Quality Sidewalk in Mountain View



Sidewalks should be set back from the street, wide enough to allow comfortable side-by-side walking, and have trees spaced close enough to provide consistent shade like this one on Evelyn Street.

Bicycle Network Coverage and Gaps

The city has a varied network of bicycle facilities, some of which are well connected and some of which are separated by gaps or barriers.

The term All Ages and Abilities (AAA) refers to bikeways designs that are comfortable for all different types of bicyclists, including the most concerned. In Access MV, the City identified a future AAA Bike Network made up of existing and planned/funded projects (Figure 12).

Facility types represented in the AAA existing and planned/funded network include but are not limited to bicycle boulevards, trails, protected bikeways and bike lanes on lower volume and speed streets (described in Figure 11).

Figure 12 shows that the density of the future AAA network varies across the city. The existing and planned/funded AAA network is robust in the northern parts of the city, including North Bayshore and Shoreline Park. The Stevens Creek Trail provides a continuous AAA-quality connection between the northern and southern parts of the City for bicyclists. In other areas, such as the residential neighborhoods in the city's southern parts, there are gaps in the AAA network, and some bikeways on major roads such as Springer Road and Grant Road may not fall under the AAA designation.

Figure 11 Types of Bicycle Facilities in Mountain View's All Ages and Abilities (AAA) Bike Network



Bicycle boulevards (Class III) are shared streets with low motor vehicle speeds and volumes with features and pavement markings designed to prioritize bicycles.



Trails (Class I) are shared-use bicycle and pedestrian paths that are separated from motor vehicle traffic. In Mountain View, most trails are paved.



Protected bikeways (Class IV) are bicycle facilities with physical barriers between bicyclists and motorized vehicles. Physical barriers can be provided with flexible plastic posts, concrete, plants and trees, or other treatments.



Bike lanes (Class II) are bicycle facilities that do not include a physical barrier separating cyclists from motor vehicle lanes.

Figure 12 Existing and Planned/Funded AAA Bicycle Facilities vs. Existing Bicycle Network

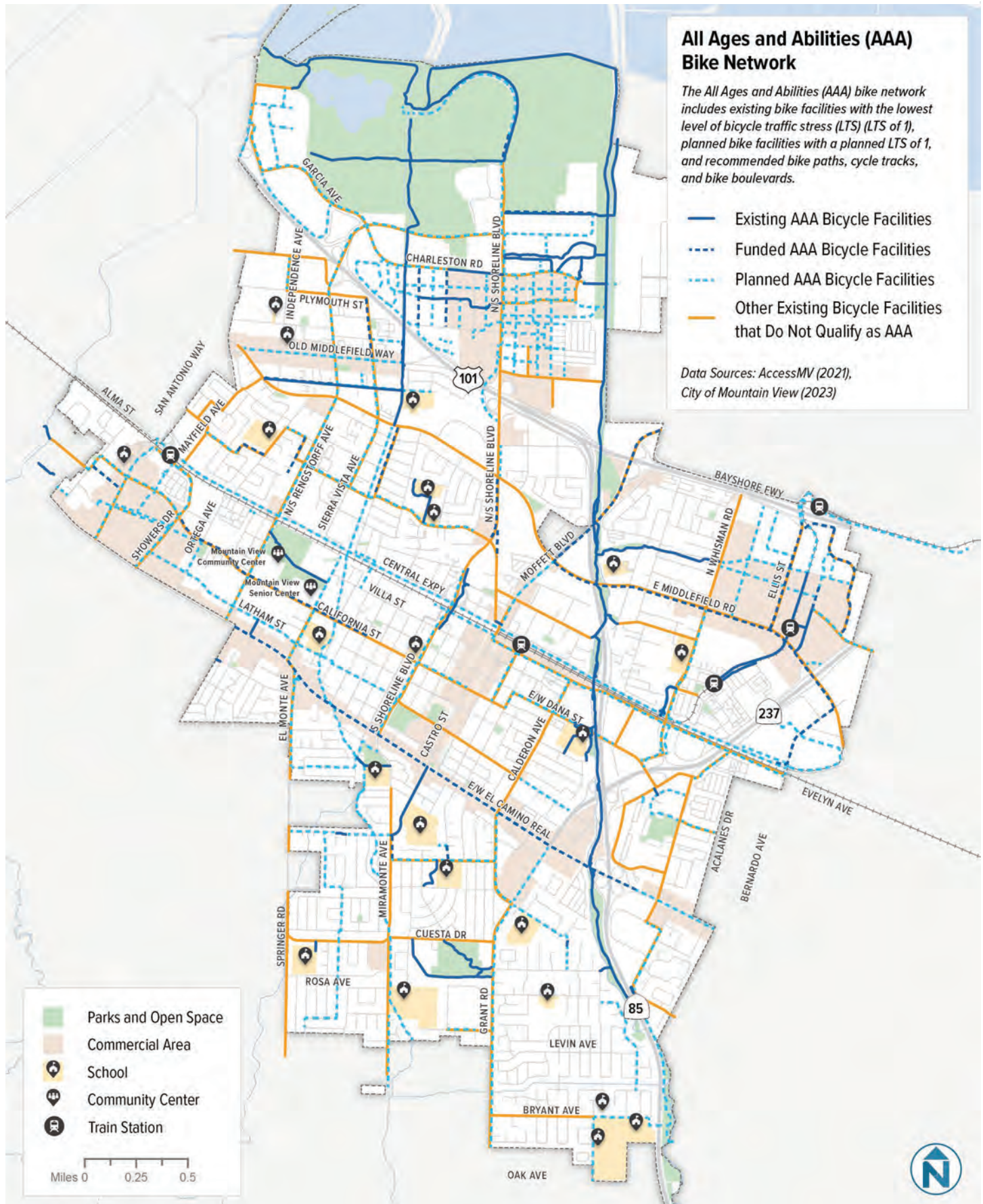


Figure 13 Bicycle Network and Challenging Routes for Walking, Biking, and Rolling

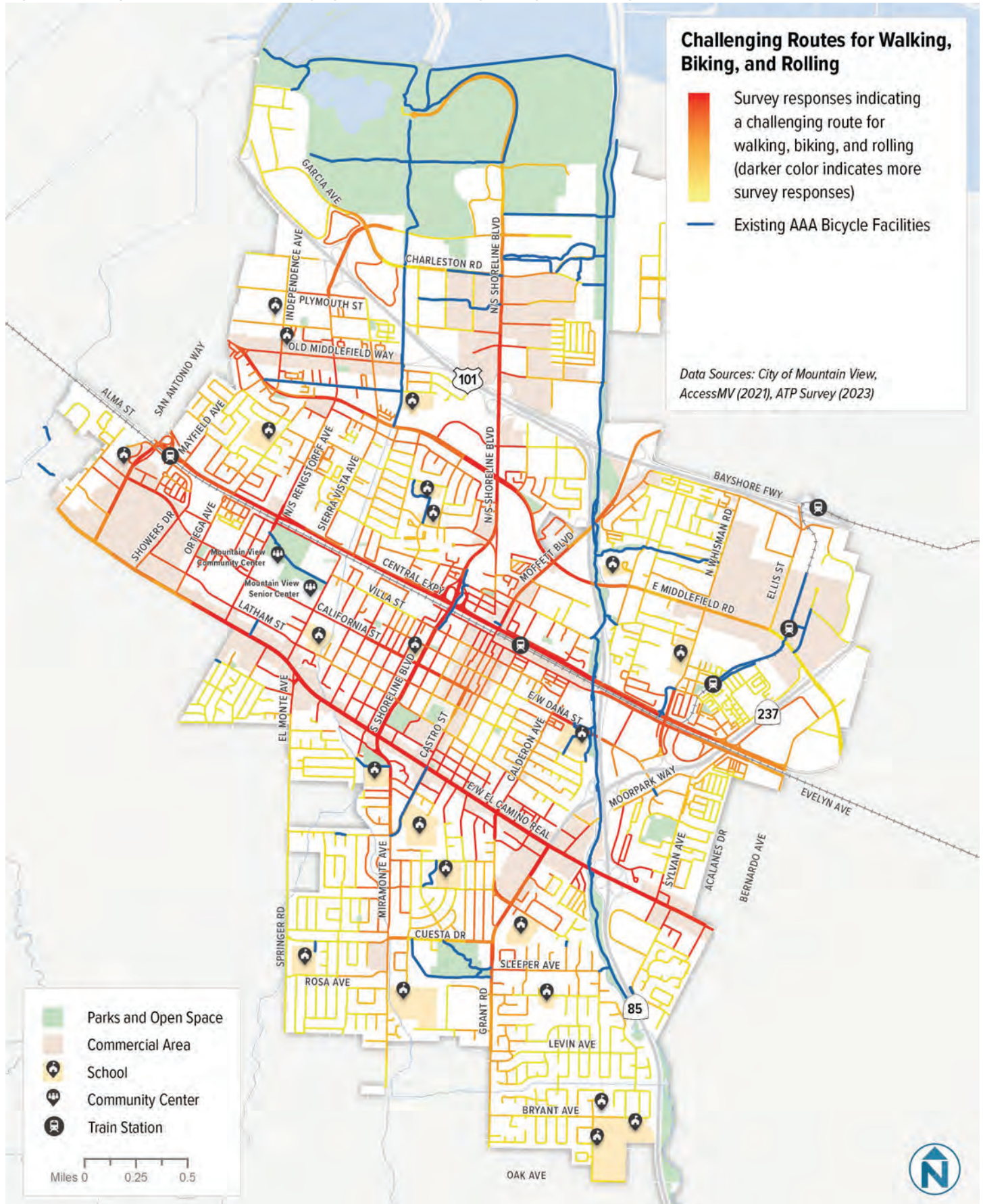
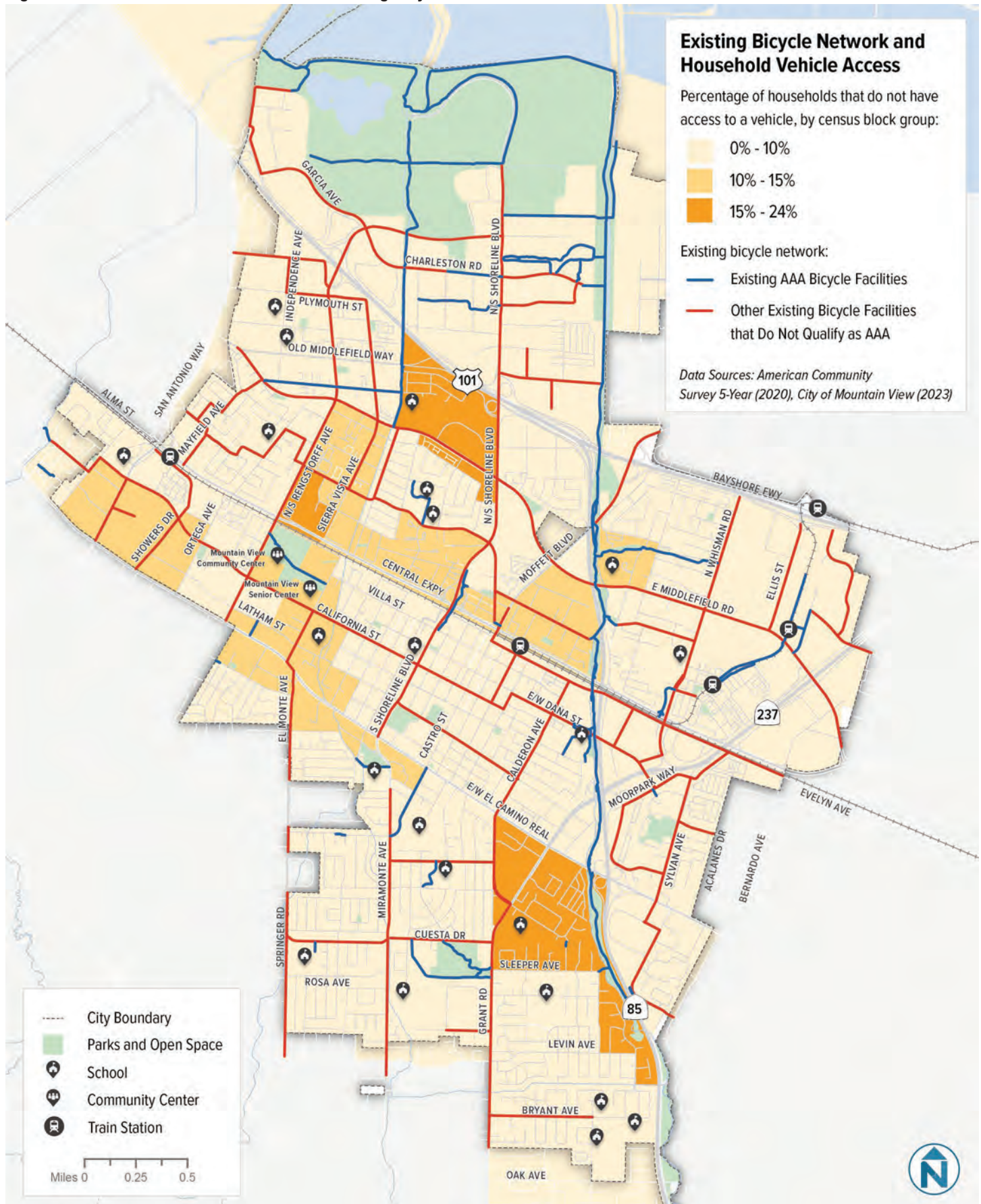


Figure 14 Household Vehicle Access and Existing Bicycle Network



Survey respondents were asked to identify challenging walking or biking routes, summarized in Figure 13 and Table 5. Note that the City does not have jurisdiction over El Camino Real or Central Expressway, these are owned by Caltrans and the County of Santa Clara, respectively.

Beginning in winter 2023, the City and Caltrans are partnering to implement bike lanes (Class II) and flex post-protected bikeways (Class IV) along El Camino between Sylvan Avenue and Rengstorff Avenue, which will significantly improve bicycle network connectivity for many community members. Community feedback about the El Camino corridor supports these improvements as well as improved bicycle access across and adjacent to El Camino Real.

Table 5 Top 10 Challenging Routes as Indicated by Community Input.

Rank	Challenging Route
1	El Camino Real* (between Dale Avenue and Rengstorff Avenue/Ortega Avenue)
2	Miramonte-Shoreline Boulevard (between Sonia Way and Wright Avenue)
3	Central Expressway* (between Thompson Avenue and SR-85)
4	Grant Road (between Bentley Square and El Camino Real)
5	Rengstorff Avenue (between El Camino Real and Montecito Avenue)
6	Latham Street (between Ortega Avenue and Leksich Avenue)
8	Escuela Avenue (between El Camino Real and Villa Street)
9	California Street (between Chiquita Avenue and Ortega Avenue)
10	Leland Avenue/Crisanto Avenue (between College Avenue and Escuela Avenue)

Source: ATP Survey (2023)

* El Camino Real is owned by Caltrans and Central Expressway is owned by the County of Santa Clara. The City does not have jurisdiction over these roads.

Bike Parking and End-of-Trip Facilities

Beyond the availability of community bikeways, a complete bicycle network includes other infrastructure and facilities that support the needs of people who bicycle. A comprehensive bicycle network includes safe and secure bike parking, accurate bicycle detection, bicycle maintenance stations, restrooms, drinking fountains, and clear wayfinding for bicyclists (Figure 15).

Figure 15 Examples of Bike Racks and Bicycle Push Button



Bike rack located in Downtown Mountain View



Bicycle push buttons allow bicyclists to request a green light

Pedestrian and Bicycle Connectivity

The City's sidewalk network provides consistent connectivity and coverage within most neighborhoods. This connectivity can be improved by providing enhanced and additional crossings of major roadways.

East/West, North/South Connections

Survey respondents were asked to identify specific points where they would like improvements. These points concentrated at intersections on major east-west roadways like El Camino Real, Central Expressway, and Highway 101 interchanges, particularly where they intersect with significant north-south roadways such as Rengstorff Avenue, Shoreline Boulevard and Castro Street/Moffett Boulevard (Figure 17). Other top locations for improved crossings suggested through survey responses and community meetings include Grant Road and El Camino Real, Escuela Avenue and El Camino Real, Showers Drive and California Street, Rengstorff Avenue and West Middlefield Road. These user-identified locations are displayed in Figure 17 and the top ten challenging spot locations identified by users are listed in Table 6. All ten of these locations are intersections or interchanges that are owned by, and under the authority of, the state or county.

Table 6 Top 10 Challenging Intersections for Walking, Biking, and Rolling as Indicated by Community Input

Rank	Challenging Spot
1	Rengstorff Avenue & Central Expressway (and by extension: Rengstorff Avenue & Leland Avenue/Crisanto Avenue)
2	Castro Street/Moffett Boulevard & Central Expressway (and by extension: Castro Street & Evelyn Avenue)
3	El Camino Real & Castro Street
4	Central Expressway & Shoreline Boulevard
5	Grant Road & El Camino Real
6	Miramonte Avenue/Shoreline Boulevard & El Camino Real
7	SR-85 & El Camino Real
8	US-101 & Shoreline Boulevard
9	SR-85 & Central Expressway
10	El Camino Real & Escuela Avenue

Source: ATP Survey (2023)

Note: The city has limited control over these locations. Central Expressway is owned and operated by Santa Clara County and El Camino Real, SR-85 and US-101 are owned and operated by Caltrans.

Distance between Crosswalks

Figure 18 shows the distance between marked crosswalks along Mountain View's major roads. Along El Camino Real and Central Expressway, some distances between crossings are more than 1,300 feet or ¼ mile. Other arterial roads with segments with long distances between marked crossings include East Middlefield Road, segments of Shoreline Boulevard, and Rengstorff Avenue.

Figure 19 shows the distance between marked or unmarked crosswalks for all the City's roads. Longer block lengths exist on several segments throughout the city, such as Showers Drive, Latham Street, California Street, East Middlefield Road, and Grant Road.

Survey respondents noted that the lack of crossings creates a significant barrier to access for pedestrians and bicyclists traveling north and south, especially when accessing downtown Mountain View from the south.

El Camino Real

Survey respondents noted that the lack of crossings on El Camino Real creates a significant barrier to access for pedestrians and bicyclists traveling north and south, especially when accessing downtown Mountain View from the south.

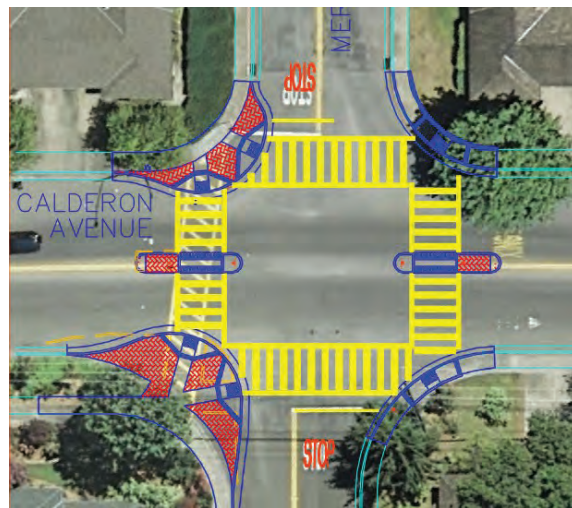
Three new Pedestrian Hybrid Beacons (PHBs) will be installed on El Camino Real at Bonita Avenue, Crestview Drive, and Pettis Avenue as part of a Caltrans repaving project that will

begin construction in 2024. These PHBs will improve connectivity and reduce the distance between intersection crossings. In addition to the three PHBs and the aforementioned bicycle improvements, plans to improve El Camino Real include curb ramp upgrades, high visibility crosswalks at all signalized intersections, and corner bulbouts at some of the unsignalized cross streets (Figure 16). Further mobility and connectivity enhancements, such as protected intersections, will be addressed through City capital projects following the Caltrans repaving work.

Figure 16 Examples of a Pedestrian Hybrid Beacon (PHB) and Curb Bulbout in Other Jurisdictions



PHBs include flashing lights and marked crossings that alert vehicles to crossing pedestrians.



Curb bulbouts reduce the distances between curbs for crossing pedestrians and reduce vehicle turning speeds.

Figure 17 Pedestrian Network and Challenging Spots for Walking, Biking, and Rolling

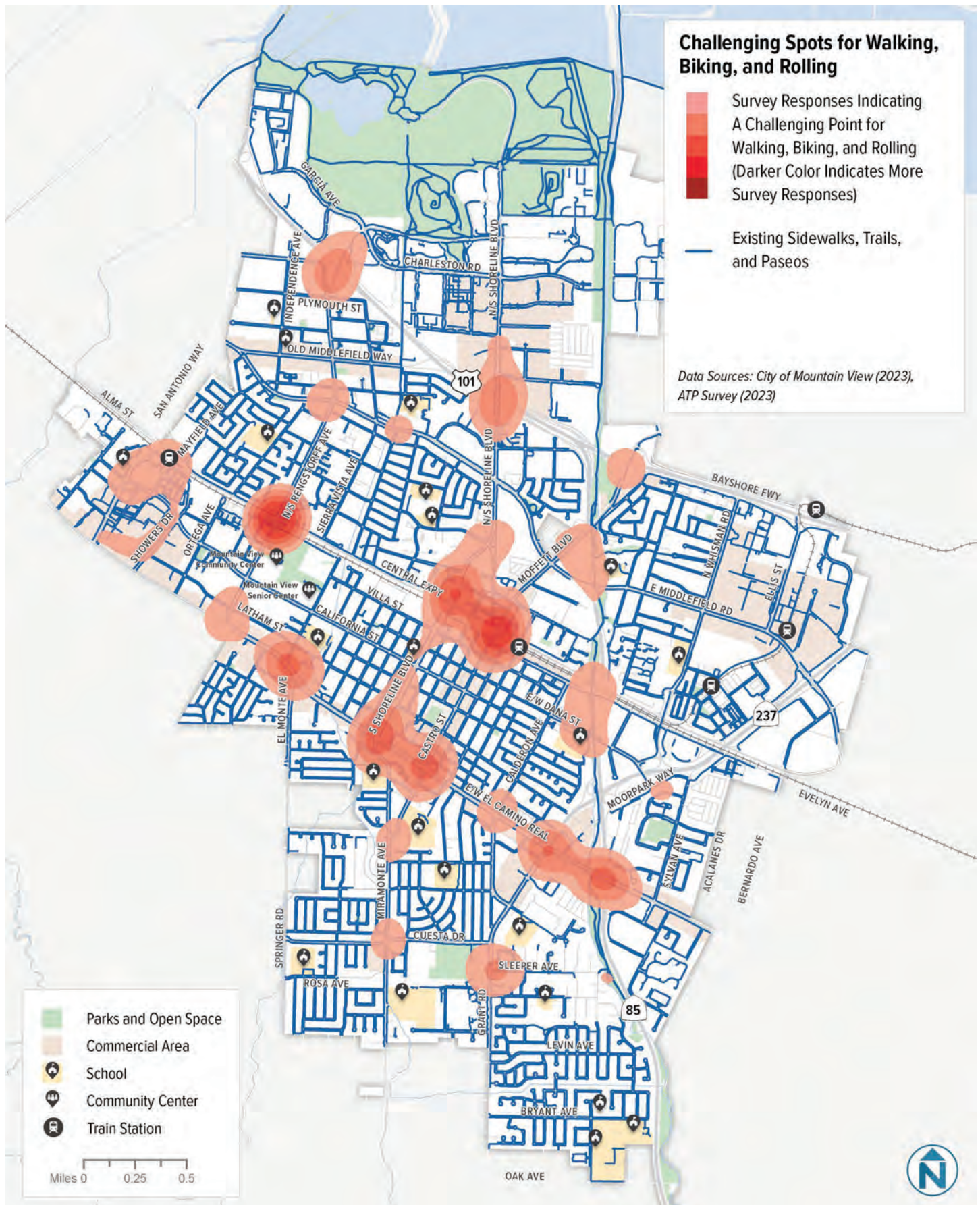


Figure 18 Distance Between Marked Crosswalks (Major Roads)

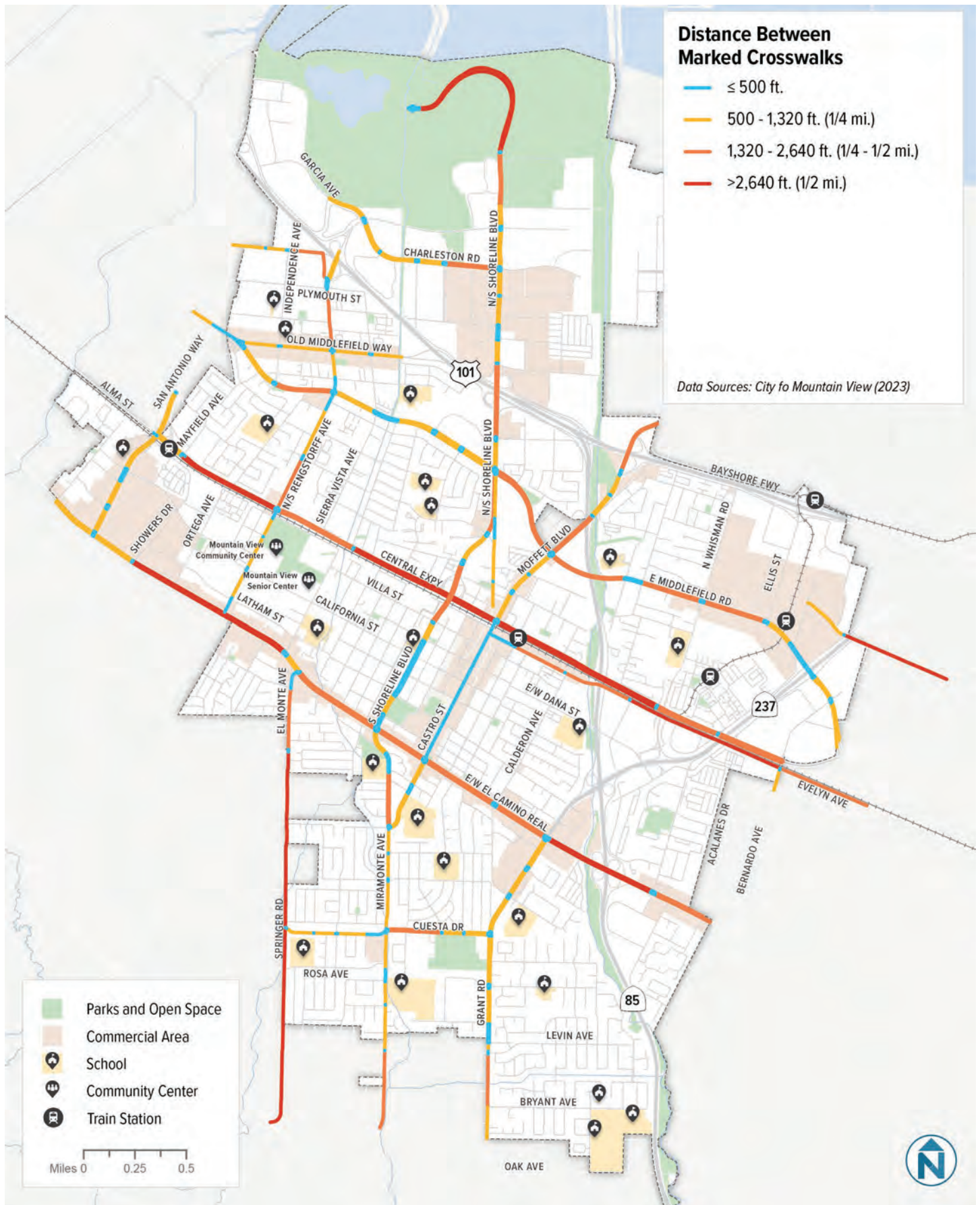
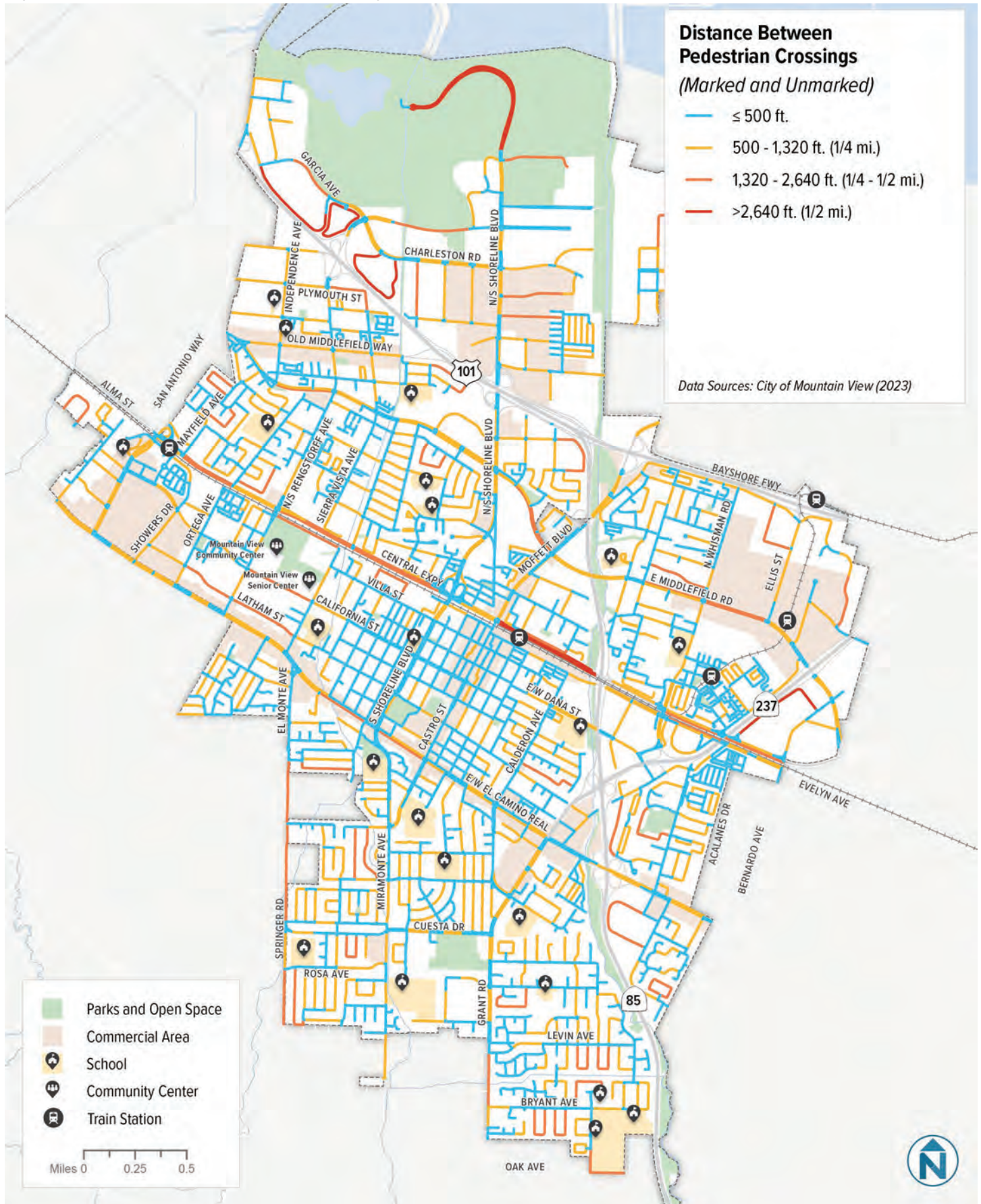


Figure 19 Distance Between Pedestrian Crossings (Marked or Unmarked, All Roads)



3. SAFETY AND COMFORT

Many places in Mountain View provide safe and enjoyable spaces for people who walk, bike, and roll—especially places where people feel separated from vehicles and offer access to nature and open space.

Community-Identified Favorite Places and Routes

Survey respondents were asked to share their favorite places to walk, bike, or roll. Many identified parks and trails in Mountain View. Trails are both routes and destinations where the car-free environments add to people’s feeling of safety and enjoyment. Survey respondents use Stevens Creek Trail for a wide variety of purposes, including recreational bike rides with family and kids, commuting to work, and accessing key destinations like grocery stores and the library. Respondents expressed appreciation for the safe pedestrian and bicycle environment of Sylvan Park and Rengstorff Park. In other places, such as Shoreline Park, respondents reported driving to these places to walk or bike.

Downtown, especially the pedestrian-only portion of Castro Street, was celebrated as a great place for walking by survey respondents and community conversation participants. The safe and enjoyable environment, the appeal of no automobile traffic, and the availability of many fun activities and destinations, make this area a favorite for community members. (Figure 20)

Figure 20 Examples of Favorite Places to Walk and Bike in Mountain View



Trails that provide a space for walking and biking away from motorized vehicles were popular destinations for survey respondents.



Survey respondents enjoy walking in the car-free segments of Castro Street downtown.

Community-Identified Challenging Locations for Walking and Biking

Survey respondents were asked to map challenging places to walk, bike, or roll. Challenging locations were concentrated along El Camino Real, at Rengstorff Avenue and Central Expressway intersection, and at crossings of US 101 and SR 85. Lastly, survey respondents and community members identified the intersection of Castro Street, Central Expressway, Moffett Boulevard, and Caltrain as a challenging area to walk and bike.

Collisions Involving Bicycles and Pedestrians

From 2015 to 2022, there were 563 reported collisions involving people walking and biking in Mountain View, including 11 fatal and 52 severe injury collisions. According to the 2024 Mountain View Local Roadway Safety Plan (LRSP), 73% of fatal and severe injury (FSI) collisions occurred along arterials with higher vehicle speeds (35 mph or above), more vehicle lanes, and higher traffic volume.

As shown in Figure 21, fatal and severe-injury (FSI) pedestrian collisions were particularly concentrated along arterial roadways, which represent 20% of the city's street network. As shown in Figure 22, FSI bicycle crashes were distributed throughout the city. However, when looking at all crashes, not just FSI crashes, the locations generally overlap with locations that community members reported as challenging in the MVATP survey. See Figure 23 and Figure 24.

Near Misses

Pedestrian and bicycle collisions are usually underreported in many communities for a variety of reasons. Near misses are also not consistently reported but provide additional information for decision makers. Figure 25 maps near miss reports submitted to AskMV between 2021 and 2022. The locations are in the Downtown area, Rengstorff Avenue, El Camino Real and neighborhood locations. Given that the data presented in Figure 25 is self-reported and limited, it should not be used to draw broad conclusions.

Figure 21 Collisions Involving People Walking by Severity of Injury and Challenging Locations

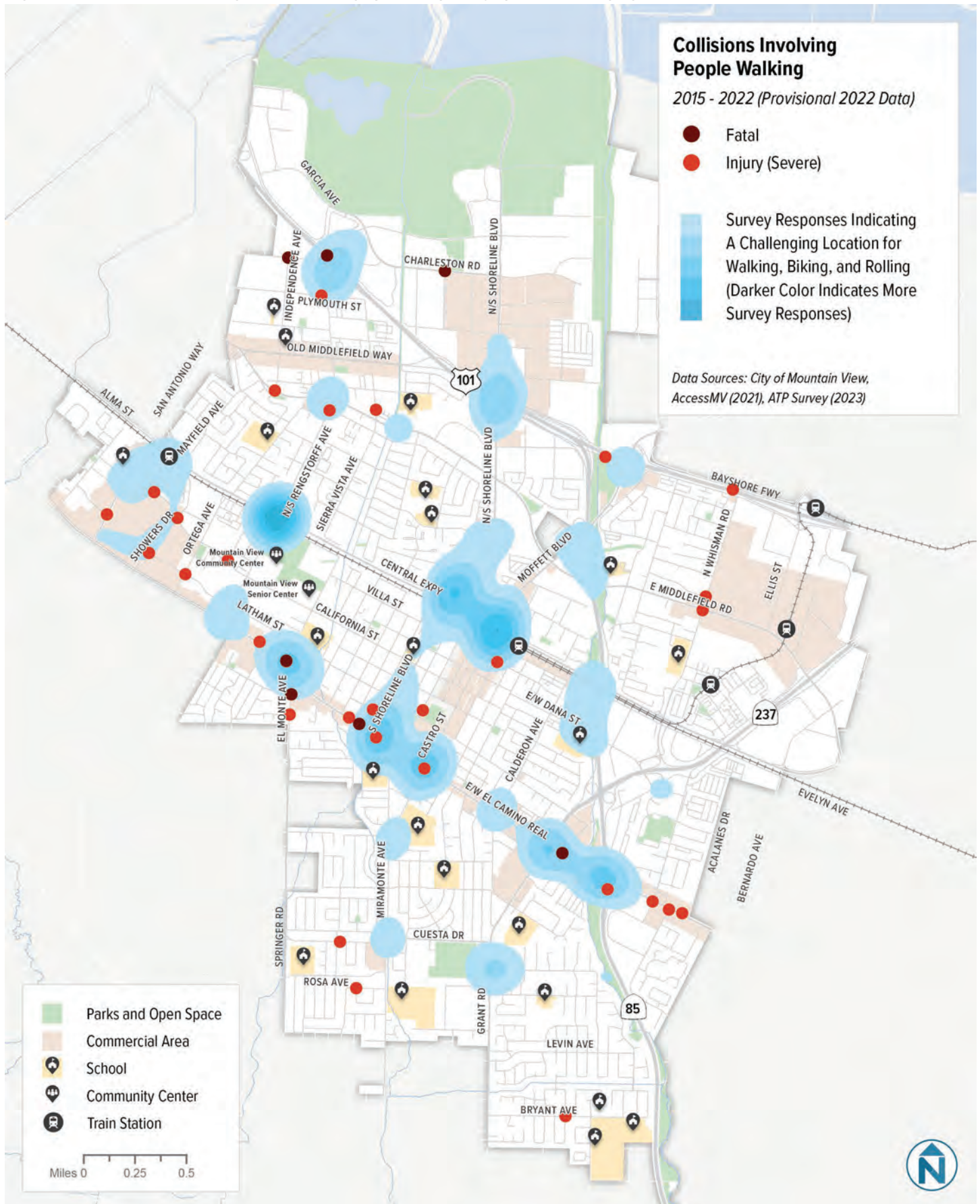


Figure 22 Collisions Involving People on Bikes by Severity of Injury and Challenging Locations

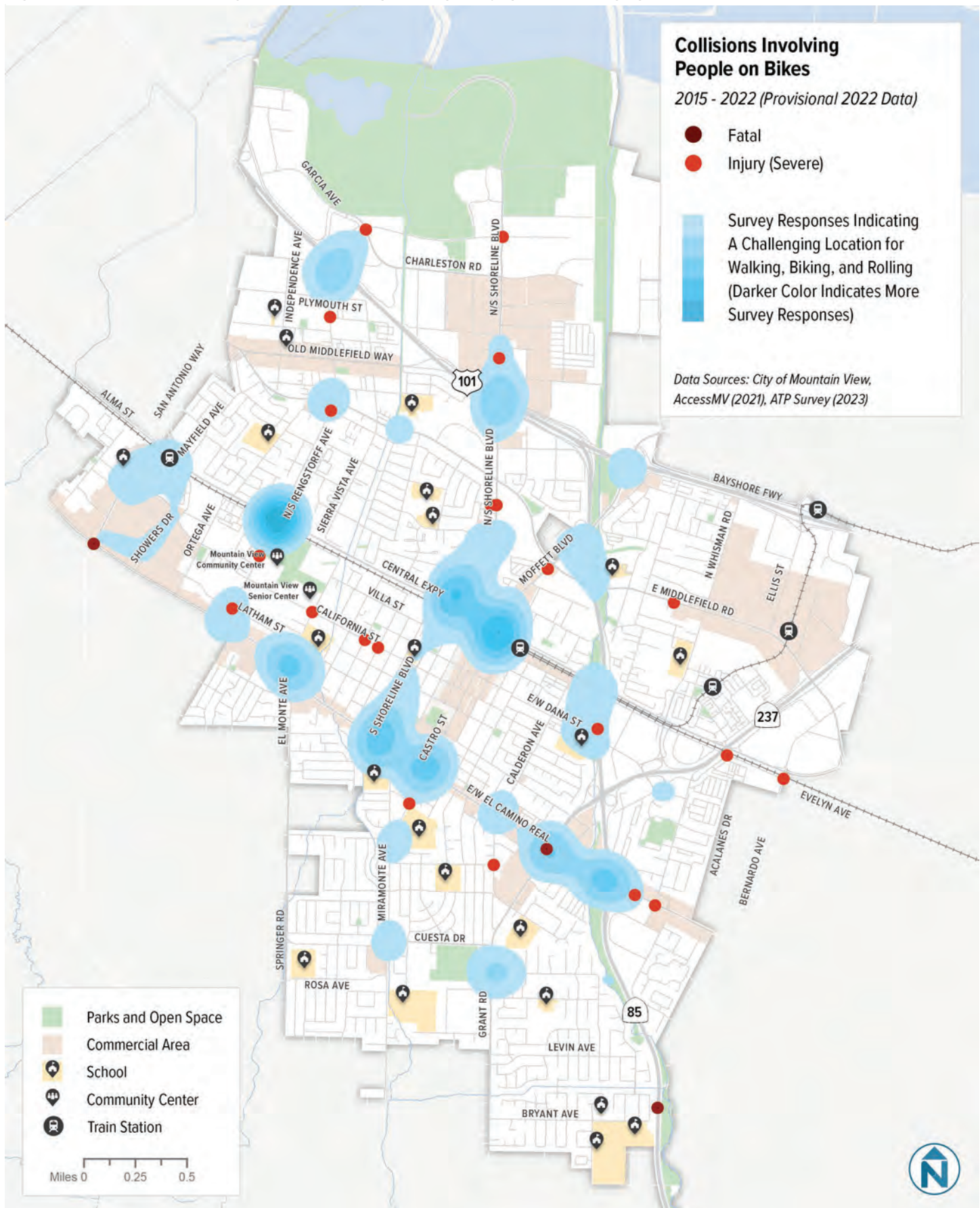


Figure 23 Frequency of Collisions Involving People on Bikes and Challenging Locations

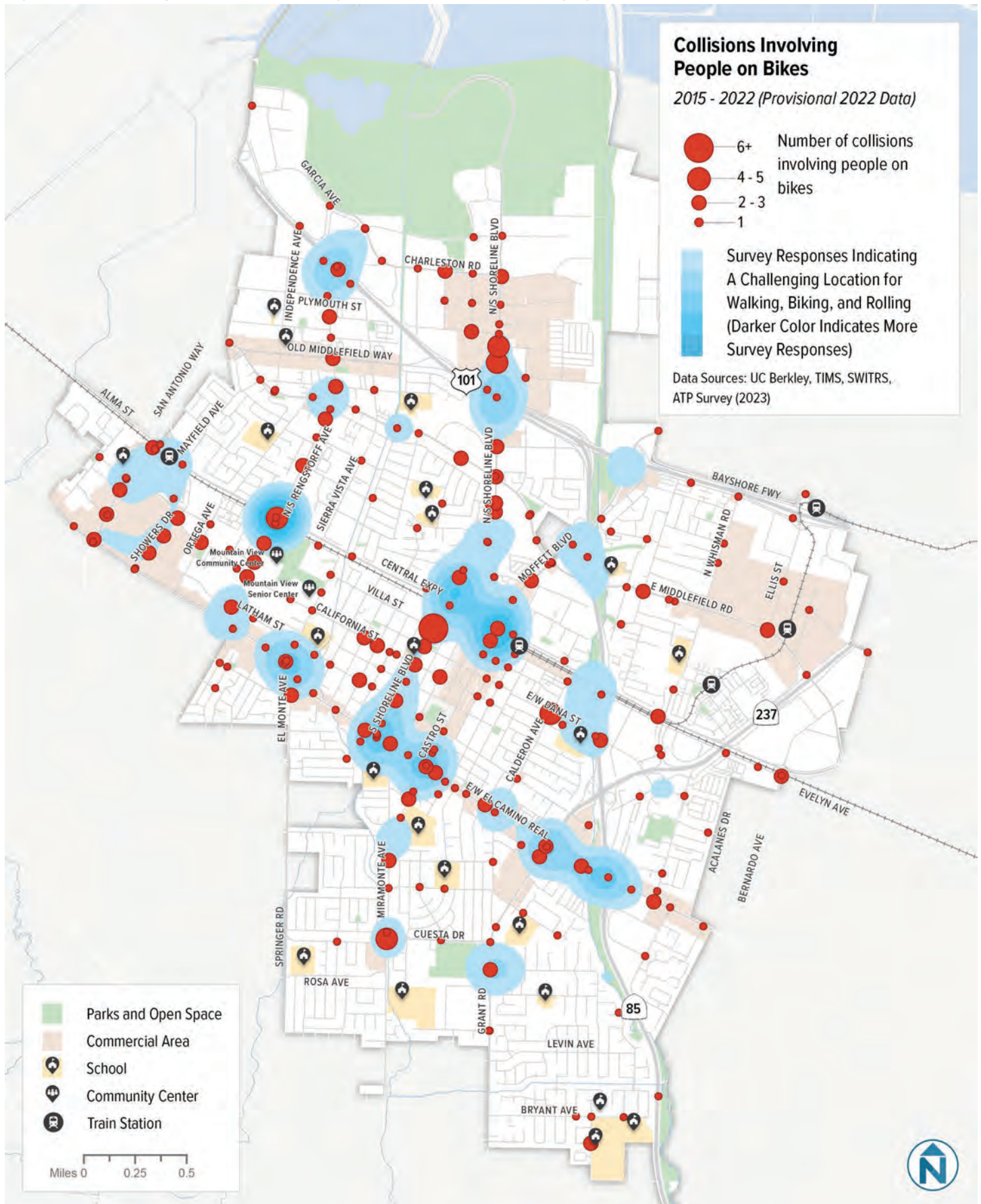


Figure 24 Frequency of Collisions Involving People Walking and Challenging Locations

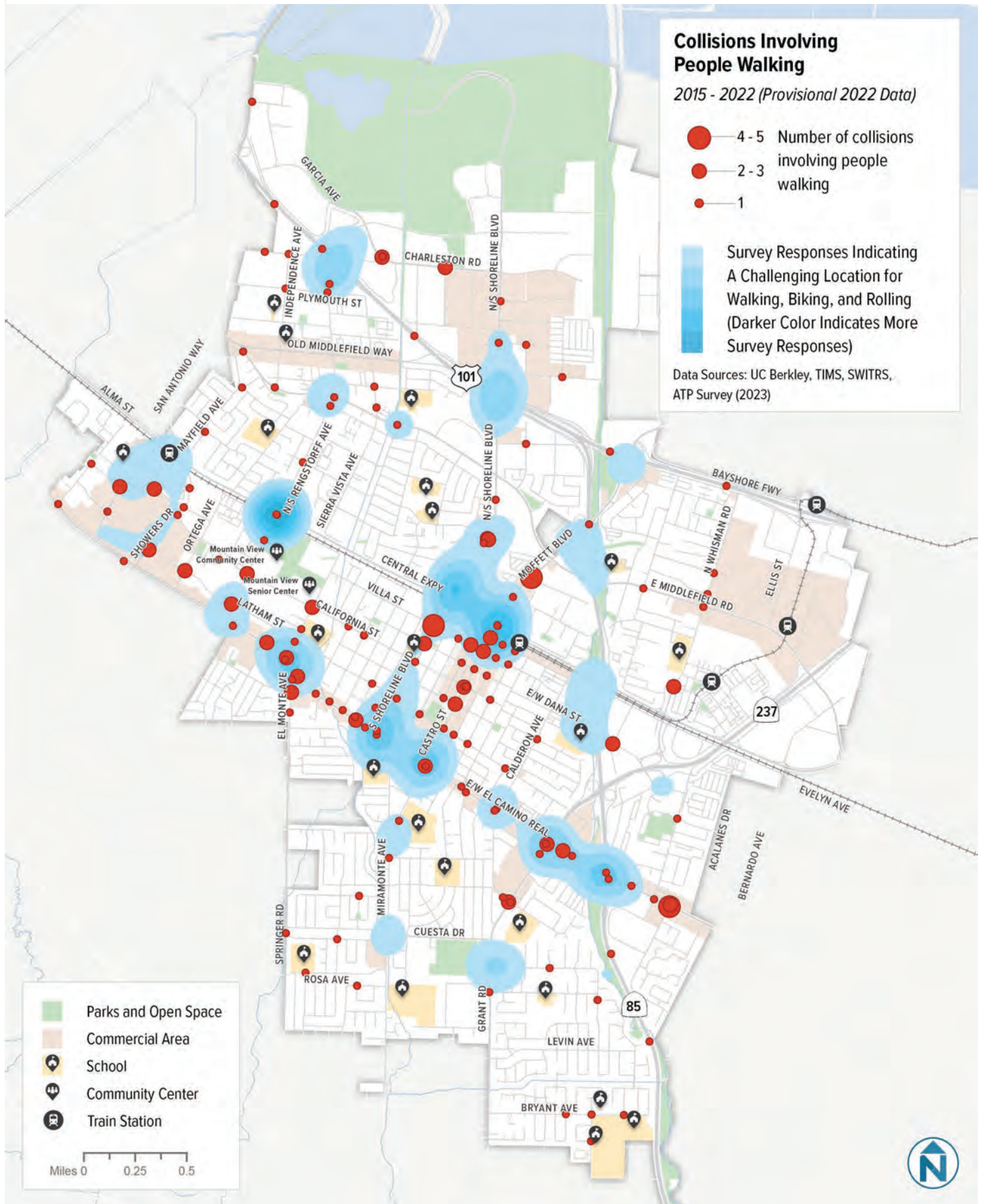
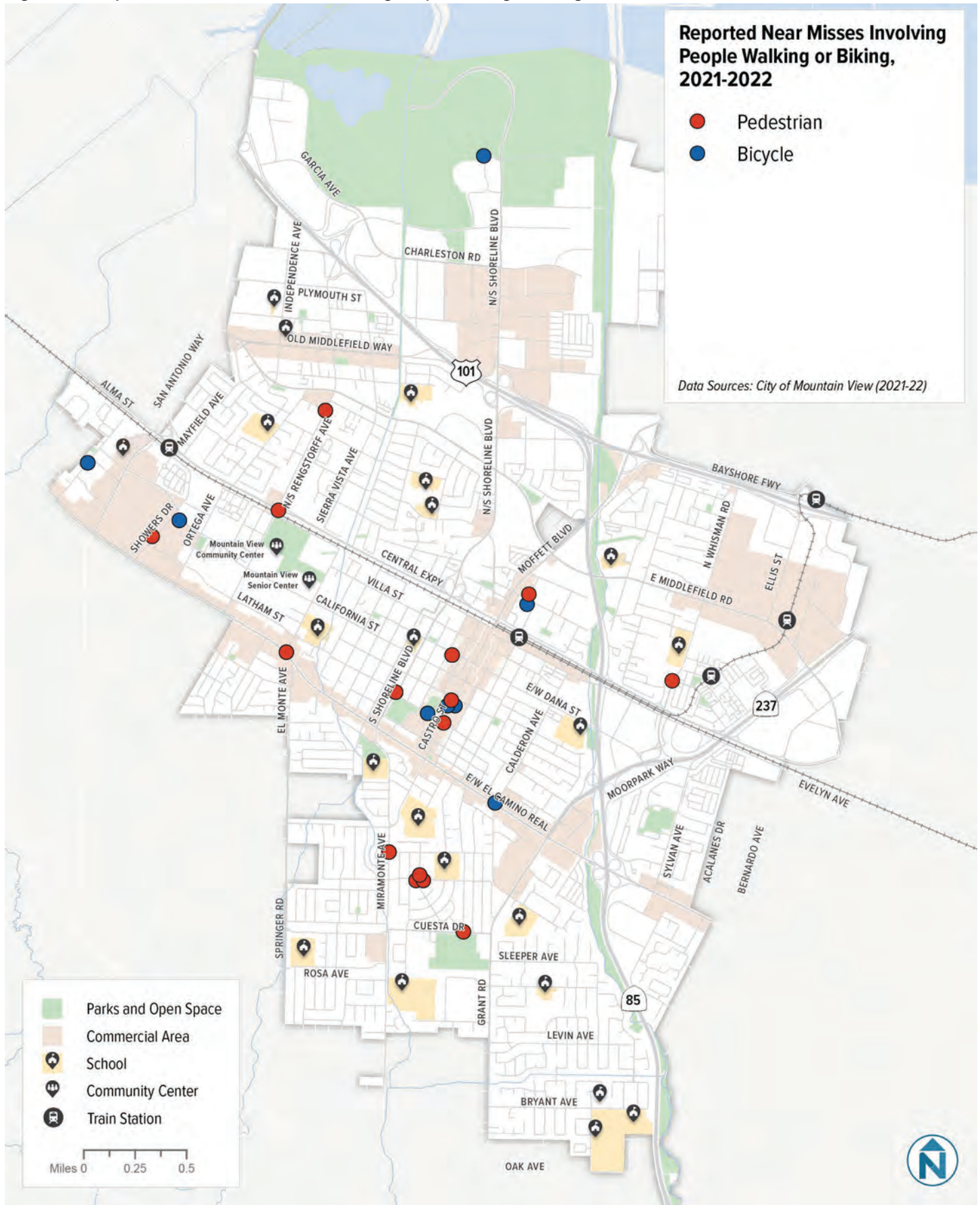


Figure 25 Reported Near-Miss Collisions Involving People Walking or Biking, 2021-2022



Sidewalk and Roadway Conditions

The City has 277.8 linear miles of sidewalk, which represents 96% sidewalk completion.

US Access Board's Public Right-of-Way Accessibility Guidelines (PROWAG) provides guidance for accessible pedestrian facilities, including but not limited to stability of surface, width, slope, cross-slope, curb ramps, and detectible surfaces. The City is currently undertaking an ADA Self-Evaluation and Transition Plan that will provide detailed data on accessibility.

Community engagement for the Active Transportation Plan also noted conditions that affect user comfort including:

- leaves, dirt, gravel, debris and drainage issues on some sidewalks, bike lanes and roadway edges;
- uneven or bumpy pavement for pedestrians or bicyclists, including at crosswalks, sidewalks, curb ramps, and places where paving materials transition;
- difficult-to-see curb edges, abrupt driveway transitions, gutter joints in bike lanes, and driveway lips, including near schools and senior facilities;
- parked cars blocking bike lanes or sidewalks with rolled curbs; and
- road segments with a high number of driveways and potential vehicle-pedestrian conflict points.

Roadway Crossings, Signals, and Visibility

The City has 2,042 crosswalks including 1,156 marked and 886 unmarked crossings, in addition to 92 signalized intersections and more than 30 LED-enhanced crosswalks or rectangular rapid flashing beacons. These crossing facilities are displayed in Figure 27.

Crossing vehicular roadways, especially those with fast-moving traffic and multiple lanes, was a recurring comfort and safety challenge as indicated by community members. While the City continues to improve and enhance crossings and intersections with features that support pedestrian and bicycle safety, such as high visibility crosswalks, raised crosswalks, RRFBs, PHBs, and "daylighting" or removing visual barriers near intersections to improve visibility, some community members expressed that driver behavior continues to create challenges for safe crossings—especially across larger arterial roads such as El Camino Real and Central Expressway. As seen in Figure 28, both of those roadways currently lack enhanced pedestrian crossing infrastructure, though the previously mentioned project on El Camino Real that will begin in spring 2024 will bring some improvements along this roadway.

In contrast, the North Bayshore area has several pedestrian signals and enhanced crosswalks.

Some survey respondents highlighted that shade is limited in some areas where there are long waits at signalized intersections.

Lastly, some survey respondents encountered visibility challenges between bicyclists and vehicles while entering or exiting trails in Mountain View.

Figure 26 Example of Enhanced and Marked Crossings in Mountain View



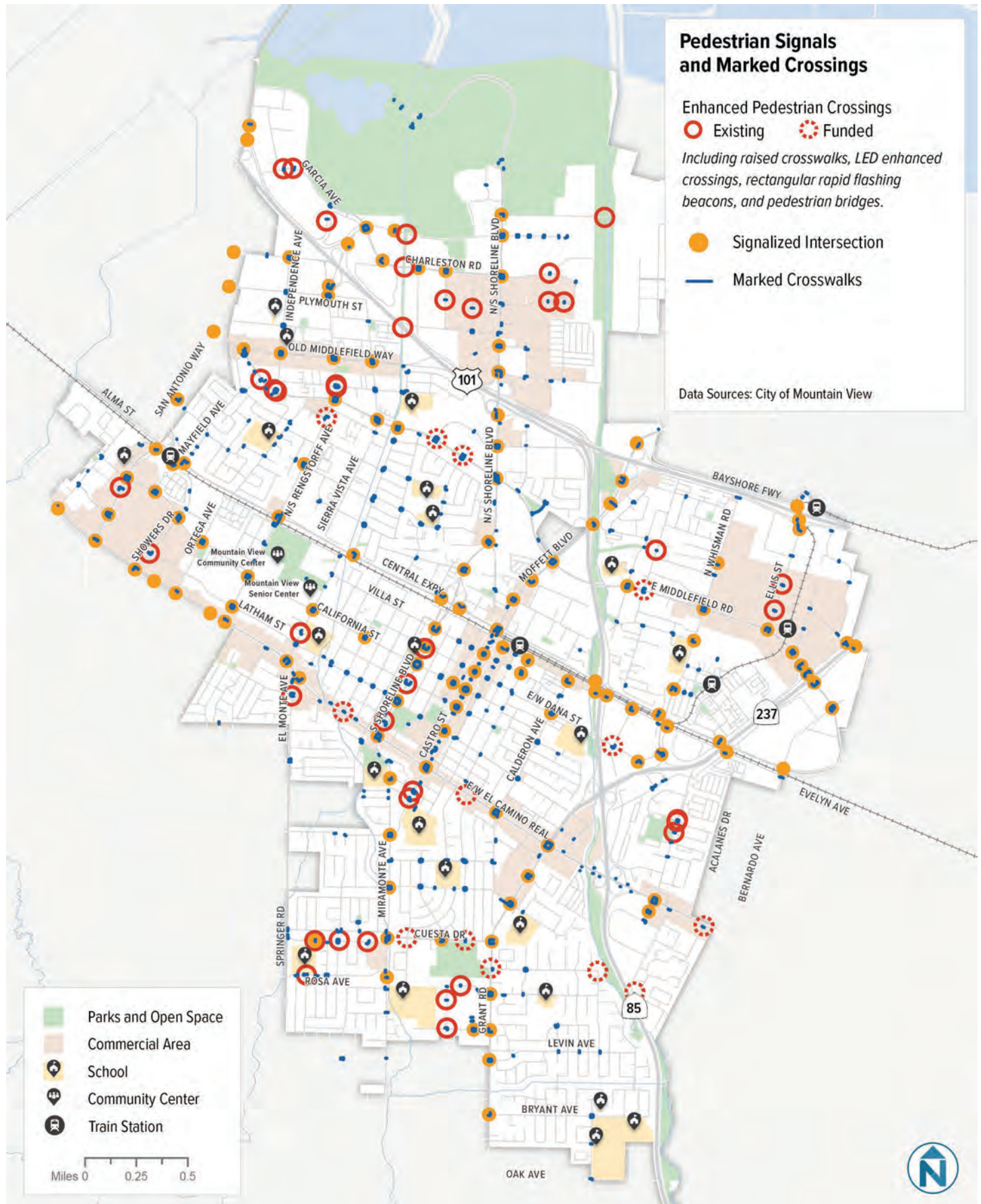
This raised crosswalk and pedestrian-activated LED enhanced sign is located near Mistral Elementary

Land Use and Urban Design Considerations

Through the ATP survey and other engagement conversations, many community members highlighted how land use and urban design can affect safety and comfort for walking, biking, and rolling. For example, in some parts of Mountain View, the shape and limited connectivity of the street network leads to lengthy pedestrian and bicycle routes to destinations that are relatively close (Figure 9 and Figure 12). However, treatments such as new mid-block crossings and paseos that can mitigate otherwise circuitous pedestrian and bicycle paths and create more shaded, pleasant ways to travel.

Survey respondents and community conversation participants frequently identified vibrant, walkable shopping areas such as downtown and the San Antonio Shopping Center as favorite places for safe and comfortable walking. As part of the AccessMV study (2021), the City analyzed Pedestrian Quality of Service (PQOS), which highlighted the impacts of the land use mix on the pedestrian experience, by utilizing WalkScore (TM) data, which prioritizes connections to certain business types and community assets.

Figure 27 Marked Crossings, Signalized Intersections, and Enhanced Intersections



4. ACCESS AND EQUITY

For many, biking, walking, and rolling are essential transportation options for meeting daily travel needs. Equitable access to key destinations can be challenging even in neighborhoods with a complete and connected pedestrian and bicycle network. An accessible and equitable active transportation network must connect people with critical services and destinations and be designed to reflect different needs and abilities. The same routes, facilities, and crossings that are easily navigable for some people can be stressful or difficult to navigate for others.

Equitable Access to Destinations and Services

The City has made strides in creating pedestrian and bicycle facilities that support mobility and provide affordable access for people of all socio-economic groups. Correspondingly, Mountain View has higher rates of walking and bicycling to work (3.2% and 5.6% respectively) compared to many neighboring cities (Sunnyvale, Cupertino, and Santa Clara) as well as the county, state and national rates (Table 7). Some peer cities including Palo Alto and Davis have higher rates of walking and/or biking to work than Mountain View.

Several neighborhoods in Mountain View have higher populations density, higher youth density, and a higher proportion of zero-vehicle households and/or low-income households. For these areas and households, gaps in the active transportation network can make biking, walking, or rolling more challenging for those who rely on them the most. (Figure 9 and Figure 14).

Participants in walking tours in these neighborhoods noted areas for improvement:

- A connected network of AAA bicycle facilities;
- Addressing sidewalk obstructions that can make it difficult to travel with strollers
- Completion of sidewalk gaps, and connections to surrounding neighborhoods and destinations;
- More shade along sidewalks

Job Access

As shown in Figure 28, the existing AAA network in Mountain View includes Stevens Creek Trail, Permanente Creek Trail, and protected bikeways along Charleston Road and S. Castro Street. These bikeways connect residential neighborhoods to many of the city's major employers, areas of high job density, and recreational areas. The planned protected bikeway on El Camino Real will further extend AAA access to jobs.

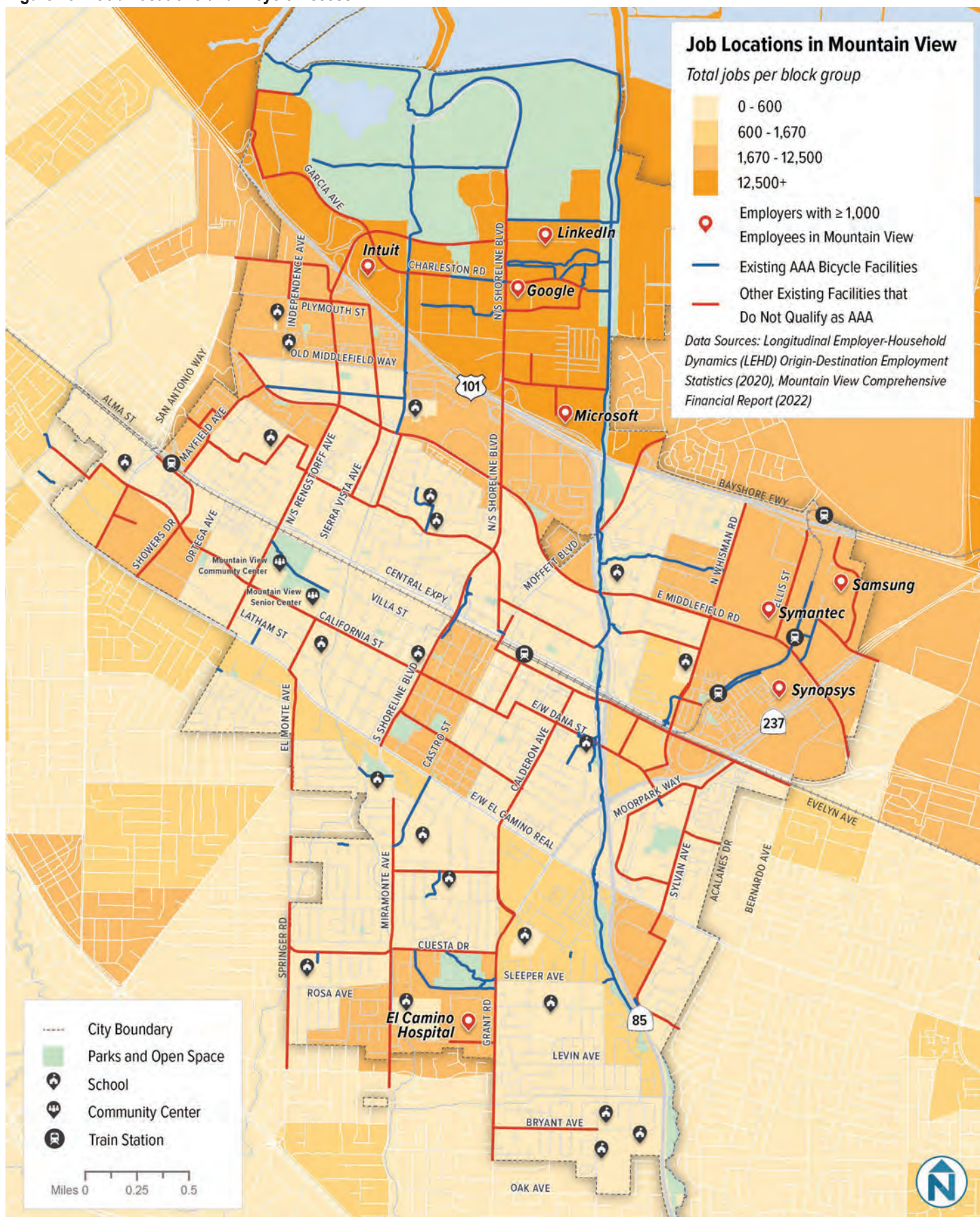
Figure 29 compares the sidewalk network to job density. The sidewalk network is nearly complete, and all job dense areas have sidewalks. The sidewalk network follows the respective roadway network, so office parks in North Bayshore have a circuitous sidewalk network, and downtown Mountain View has a dense, grid network.

Table 7 Commute Mode Share in Mountain View and Peer Cities (2021)

	Total Workers (Age 16+)	Drove alone	Carpooled	Public transit	Walked	Bicycle	Taxicab, motorcycle, or other	Worked from home
Mountain View	44,414	53.20%	4.50%	2.10%	3.20%	5.60%	1.20%	30.20%
Sunnyvale	87,798	61.6%	9.1%	5.1%	1.6%	1.6%	1.5%	19.6%
Cupertino	27,965	67.7%	5.8%	1.8%	2.8%	0.5%	1.4%	20.0%
Santa Clara (City)	69,840	58.2%	5.7%	2.2%	2.5%	0.9%	2.3%	28.3%
Palo Alto	33,763	46.10%	2.40%	1.60%	6.40%	9.20%	1.70%	32.60%
Davis	31,129	51.90%	5.20%	7.10%	3.00%	13.80%	1.00%	17.90%
Santa Clara County	967,786	59.3%	8.9%	2.1%	2.0%	1.5%	1.5%	24.8%
California	18,283,118	70.10%	9.60%	4.10%	2.40%	0.80%	1.60%	11.40%
United States	160,577,736	73.20%	8.60%	4.20%	2.50%	0.50%	1.40%	9.70%

Source: ACS 2021 5 Year Estimate

Figure 28 Job Locations and Bicycle Access



Job Locations in Mountain View

Total jobs per block group

- 0 - 600
- 600 - 1,670
- 1,670 - 12,500
- 12,500+

Employers with $\geq 1,000$ Employees in Mountain View

Sidewalks, Trails, and Paseos

Data Sources: Longitudinal Employer-Household Dynamics (LEHD) Origin-Destination Employment Statistics (2020), Mountain View Comprehensive Financial Report (2022)

City Boundary

Parks and Open Space

School

Community Center

Train Station

Miles 0 0.25 0.5

Equitable Access to Transit

Fixed-route transit services in the City include VTA light rail, Caltrain service, VTA bus service, and Mountain View shuttle service. While some community members have attractive and reliable pedestrian and bicycle access to these services, others report challenges or barriers to access.

VTA light rail station areas like Whisman Station are well connected by a complete sidewalk network and low-stress bicycle routes. In some areas, such as Pacific Drive near Whisman station, station area planning has helped create walkable, shaded streets that improve transit access for people who walk, bike, and roll. Some survey respondents identified access to a transit stop or station as a favorite destination for walking, biking, and rolling, especially the downtown transit center and other rail stations. Other respondents reported challenges getting to and from transit, especially bus stops on busy roadways with widely spaced crossings (Figure 30).

In most of the city, bus services on major roads and arterials are connected to adjacent neighborhoods via low-stress bicycle and pedestrian routes. However, most major roads and arterials (on which buses operate) can be more stressful for bicyclists and pedestrians and can make the first and the last mile access challenging. Some of the highest transit ridership in Mountain View occurs along El Camino Real, which is not a comfortable environment for many pedestrians or bicyclists, has long block lengths and crossings, and sees a higher concentration of crashes (Figure 31).

Figure 30 Mountain View Transit Center as a Destination



Community members identified the Mountain View Transit Center as a favorite destination to walk, bike, or roll to.

Figure 31 Transit Ridership and Bicycle Access

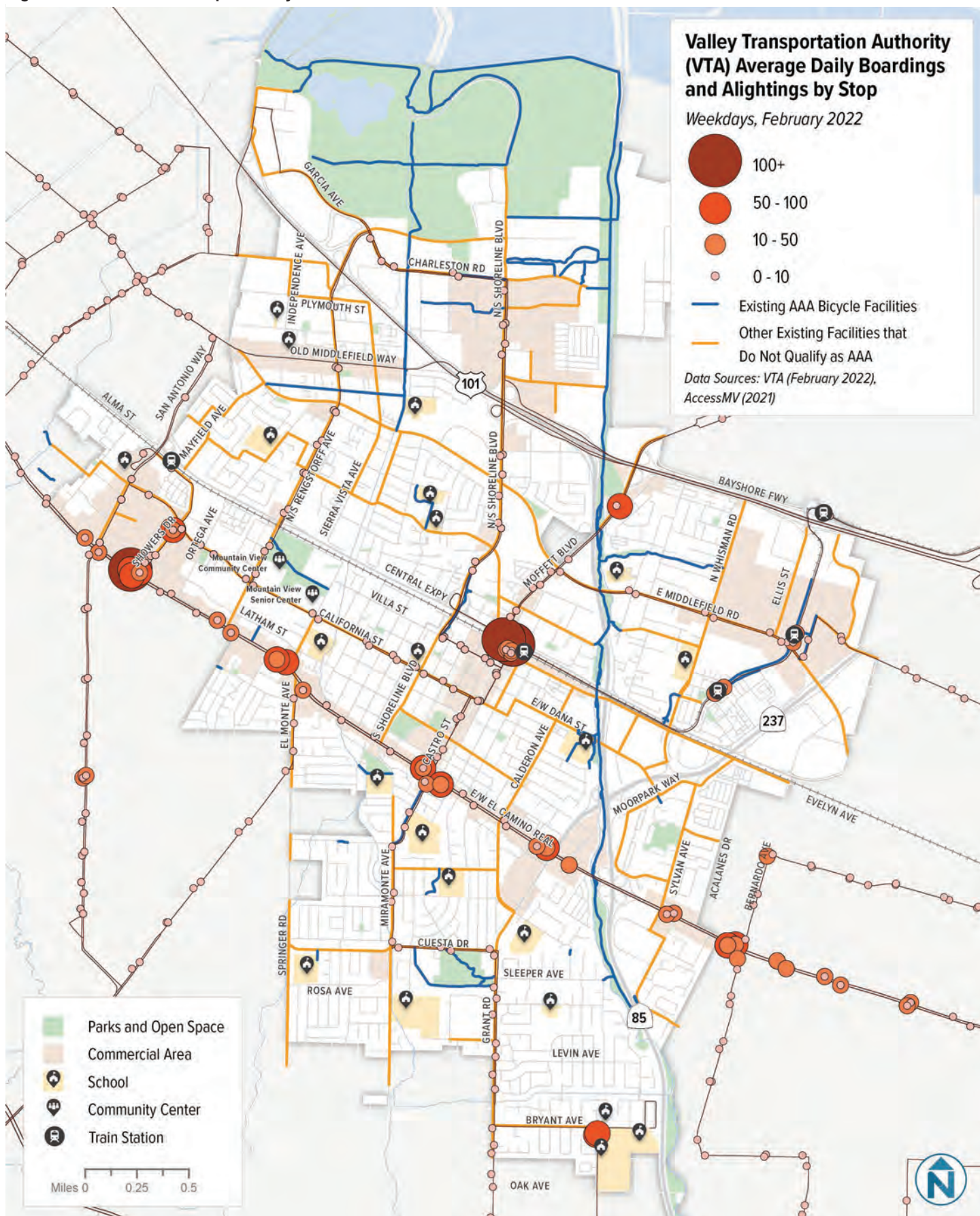
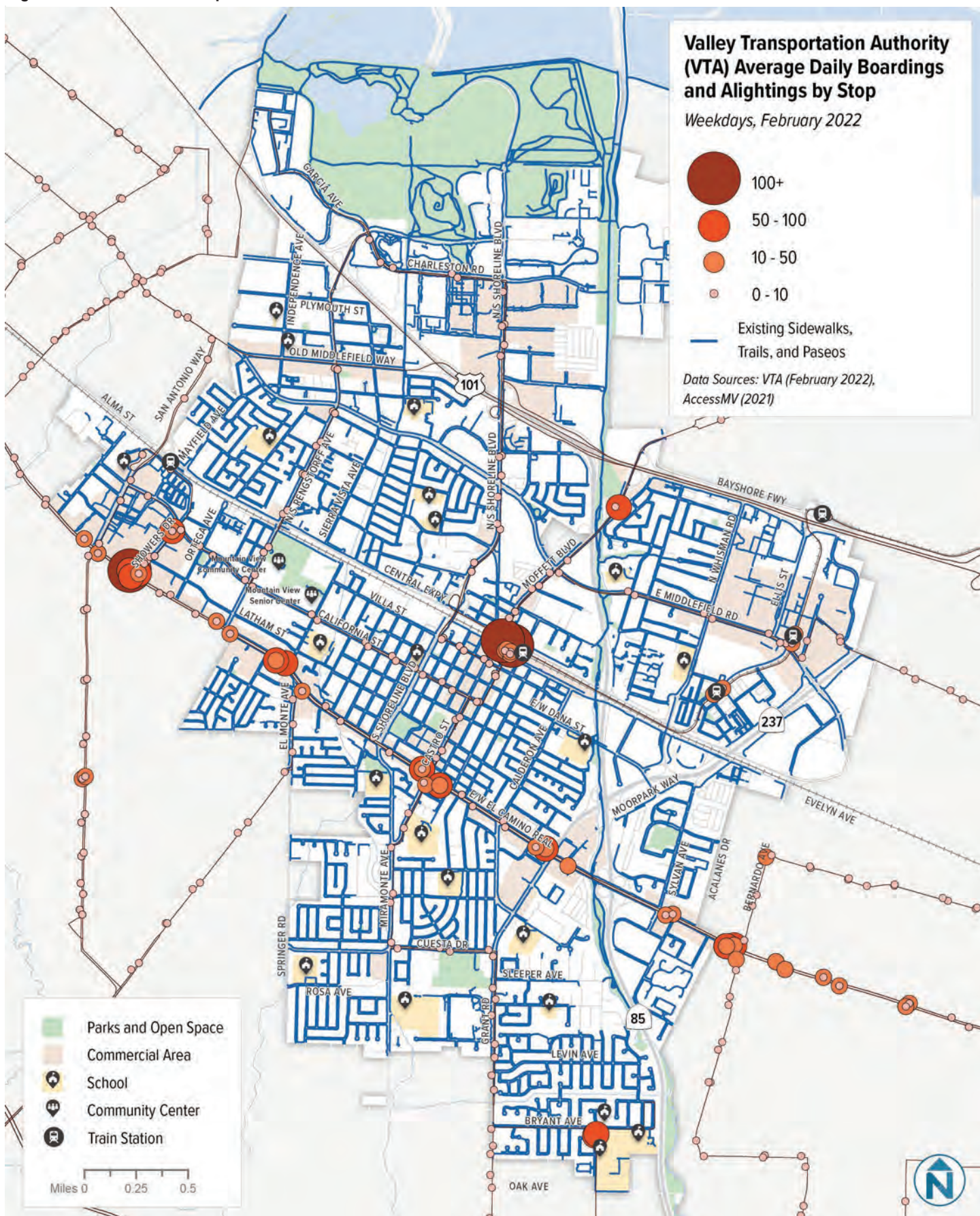


Figure 32 Transit Ridership and Pedestrian Access



5. SUSTAINABILITY AND BIODIVERSITY

In 2021, the Mountain View City Council established a goal to “protect and enhance local ecosystems and biodiversity through rewilding and other measures” as part of their Sustainability & Climate Resiliency strategic priority. The City is now undertaking a biodiversity strategy to enhance the health of local ecosystems and prevent species loss through a multi-pronged approach. One element of this approach is planning for active transportation networks that double as wildlife corridors to preserve biodiversity assets, create new spaces for local native plants, and support local ecosystems.

Defining Biodiversity

Biodiversity refers to the variety of living organisms on Earth.

California and the Bay Area are global biodiversity hotspots because they contain a high number of native species including plants, animals, birds, bacteria, fungi and insects that are found nowhere else in the world! These species do not just live in wilderness areas a long way away from cities; they also include native plants that grow (or could grow) along public streets, trails and parks, as well as the birds, insects and other living organisms that depend upon these plants.

Native plants evolved to thrive in the soils, moisture and changing weather conditions of this region. They therefore require less supplemental watering and pest control than many non-native species and may not perform well with watering regimes designed for other species.

Additionally, native species coevolved to have specific symbiotic relationships with other living organisms. For example, dwarf plantain (*Plantago erecta*) is the primary host plant to the endangered Bay Checkerspot Butterfly; showy milkweed (*Asclepias speciosa*) is a host plant for the Monarch Butterfly; and Coast Live Oak (*Quercus agrifolia*) is a keystone species which serves many other species within the ecosystem.

In areas like Mountain View, a loss of biodiversity is associated with many factors including habitat destruction, degradation or fragmentation for development, introduction of invasive species, non-native species, monoculture, pollution, biocides, and climate change. Loss of biodiversity can result in extirpation or even extinction of species, which can have cascading effects on the wider ecosystem.

Defining Green Streets

Green Streets are streets that provide space for natural elements to support biodiversity, incorporate sustainable stormwater strategies, slow motor vehicle traffic, and prioritize active transportation modes (biking, walking, rolling) for people of all ages and abilities. Green streets are a holistic approach that combines active transportation, biodiversity promotion, and green infrastructure development.

Green Streets = Active Transportation + Biodiversity + Green Infrastructure.

Green streets are an important component within the active transportation system and can provide access to recreation, nature and open spaces through green treatments that are integrated into the street. They present a unique opportunity to support Mountain View's sustainability and biodiversity goals by providing space for natural systems to thrive within the public right-of-way. Green streets influence how people use the street, often slowing motor vehicle traffic and increasing the comfort of pedestrians and cyclists by creating space and incorporating natural elements into the street.

Benefits of Green Streets

- Can help the city comply with the Municipal Regional Stormwater Permit 3.0, which requires green stormwater infrastructure in certain transportation projects.
- Green streets support active transportation by creating healthy settings where people want to walk and bike, fostering social connections and an active lifestyle.
- Trees, vegetation and green stormwater infrastructure provide and complement traffic-calming efforts, contributing to safer streets by slowing vehicular traffic.
- Green street features make public spaces more attractive and comfortable for people through shade, heat absorption, visual vibrancy and clear local identity.
- Street trees and impermeable surfaces reduce heat island effects, making streets cooler and safer during heat events, and mitigating impacts of climate change.
- Green stormwater infrastructure and plants manage stormwater by increasing pervious surfaces, reducing, slowing and treating runoff, and improving water quality.
- Green streets enhance air quality by filtering out pollutants.
- Green streets with native species provide habitat corridors that support survival of native plants, pollinators and ecosystems in this region which is a biodiversity hotspot.

Challenges of Green Streets

Green streets also come with potential challenges, which can be reduced or eliminated by selecting the right tree for the right place and by timely, species-appropriate maintenance. The following potential challenges should be considered when designing a green street.

- Tree roots can uplift sidewalks, creating uneven walking surfaces.
- Property owners may incur costs related to planting strip maintenance and care.
- Certain trees and plants may be perceived as messier than others, dropping leaves, nuts, and fruits on the ground.
- Plants may trigger seasonal allergies especially when only male trees or plants are planted in order to avoid seeds, pods and fruit.
- Trees can be susceptible to storm damage, dropping debris and limbs, or falling over completely during large storm events.

- Trees and large shrubs can encroach on sightlines if not properly placed.

Below is a quick reference to the benefits and challenges of green streets.

Table 8 Green Streets Benefits and Challenges

Benefits	Challenges
Encourages walking and biking	Roots can uplift sidewalks.
Fosters social connections and community	Can incur cost to the property owner
Enhances air quality, filters air pollutants	Can produce debris such as leaves, nuts, fruit
Reduces temperatures, keeps streets cool	Requires maintenance
Reduces stormwater runoff and water pollution	Can exacerbate seasonal allergies
Slows vehicles, calms traffic calming	Can result in property damage if limbs fall
Increases property values for property owners	Can reduce visibility in right of way
Provides habitat for wildlife and pollinators	
Provides beauty, seasonal interest, mental health benefits	
Helps conserve energy	
Supports compliance with MRP 3.0	

Mountain View's Urban Forest

Trees are a valuable community resource and an essential component of civic infrastructure. The City of Mountain View has been committed to preserving and enhancing its urban forest since 1960 when the City Council established tree care services.

Commitment to the Urban Forest

The City of Mountain View boasts a history of dedication to preserving and enhancing its urban forest, spanning several decades. This commitment is evidenced by a series of pivotal milestones:

- 1961 - Enactment of the Street Tree Ordinance
- 1975 - Adoption of the Heritage Tree Ordinance
- 1983 - Introduction of the First Street Tree Policy and Reforestation Plan
- 1993 - Establishment of the Urban Forestry Management Plan
- 2001 - Recognition of Mountain View as a Tree City USA
- 2013-2015 - City Council goals to improve the urban forest and preserve habitat
- 2015 - Launch of the Community Tree Master Plan (CTMP)
- 2021-2023, 2023-2025 - City Council Strategic Priority to develop a Biodiversity Strategy

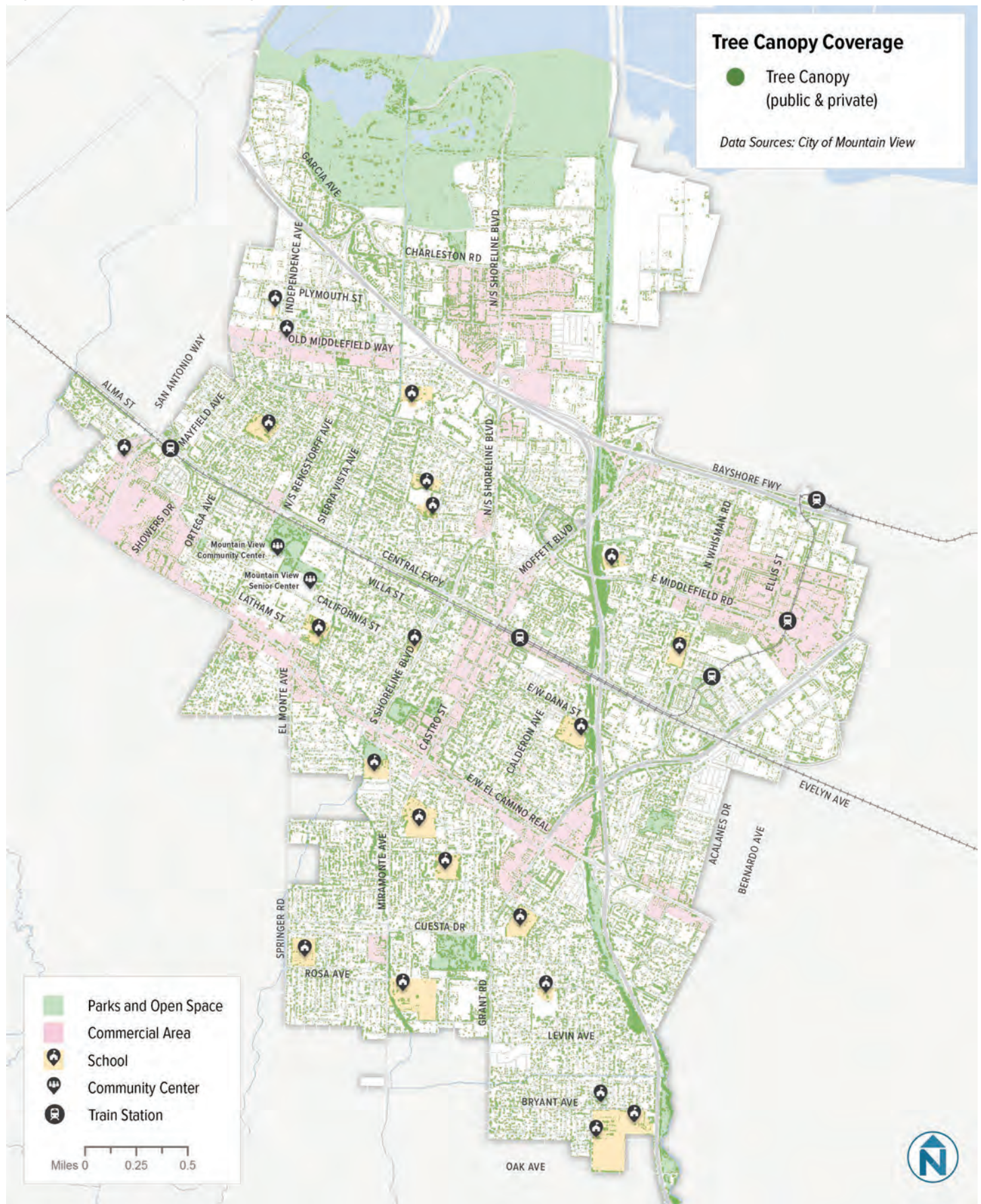
- 2022 - Revision of the Tree Technical Manual
- 2026 - Adoption of the Biodiversity Strategy and Urban Forest Plan

Tree Coverage

According to the 2022 revised *Tree Technical Manual*, Mountain View's existing urban forest covers 17.7 percent of the city with tree canopy. To put this in context, this coverage is slightly higher than the California urban area average of 15 percent (according to the USDA Forest Service) and approximately 50 percent lower than adjacent communities directly to the north of Mountain View.

A higher percentage of tree coverage brings substantial benefits, such as filtering out air pollution, reducing stormwater runoff, reducing water pollution, sequestering carbon, reducing urban heat island effects, reducing health risks associated with heat events, contributing to mental and physical health, and increasing climate resilience. In Mountain View, the 2015 Community Tree Master Plan (CTMP) establishes a goal to increase tree coverage to 22.7 percent, representing a 28 percent or 5 percentage point increase. This increase is equivalent to planting an additional 11,000 trees citywide.

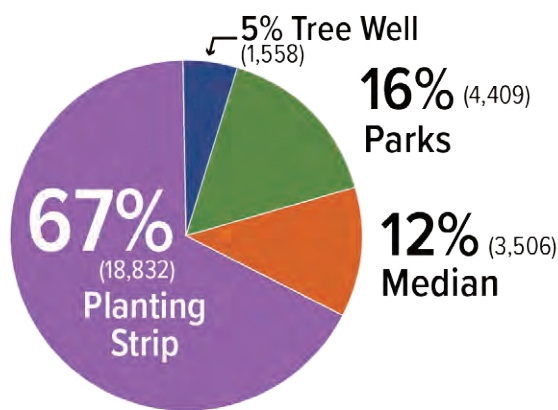
Figure 33 Tree Canopy Coverage



Street Trees in Mountain View

Mountain View has over 28,000 public trees, which enhance the City's natural beauty and provide shade and clean air. These trees are cataloged in the City's Urban Forest GIS database, encompassing attributes such as common name, species, tree group (deciduous, evergreen, palm), trunk diameter, circumference, planting type (the type of public land where the tree was planted), heritage designation, and ownership and maintenance data. According to the *Tree Technical Manual*, a significant portion of all urban trees—approximately 25%—line city streets. As shown in Figure 39, approximately 2/3 of public trees are found in planting strips, which represent an opportunity for additional community trees.

Figure 34 Public Tree Planting Types



In addition to efforts to increase permeable space and reduce the use of biocides, Mountain View's commitment to biodiversity requires a shift away from monocultural planting, which is more susceptible to diseases, and a focus on native plant species needed to support local pollinators and ecosystems.

In relation to monoculture, the City has 237 unique species of street trees, ranging from large shade trees to small ornamental trees, palms, to evergreens. While this seems like a lot of species, there are 6,000 species endemic to California. Historically, the City planted a single tree species along each street to create a unified aesthetic for the neighborhood. Single species planting creates vulnerabilities for plant health, and the City is shifting away from this type of monocultural planting.

In relation to the focus on native species, Figure 39 lists the top 10 most common street trees found in Mountain View. Of this list, Coast Redwood (*Sequoia sempervirens*) is native to California. The City has shifted toward native species for plant lists.

Figure 35 Mountain View's Top 10 Street Trees

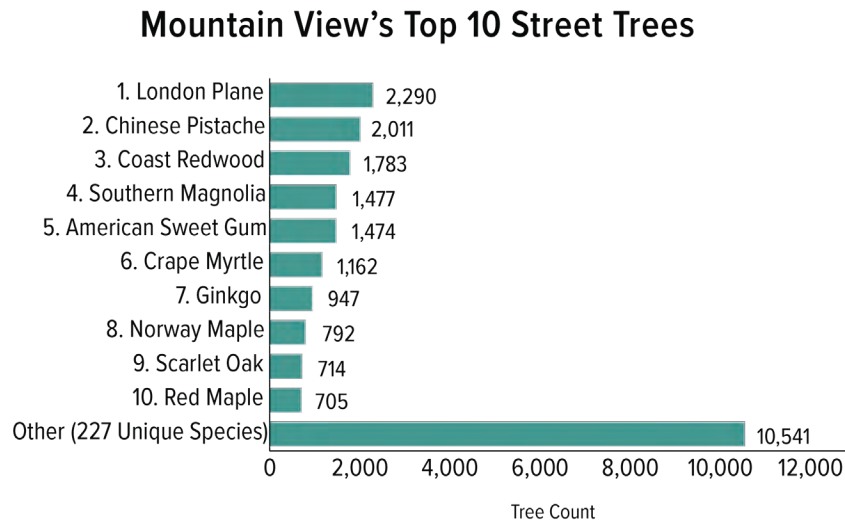
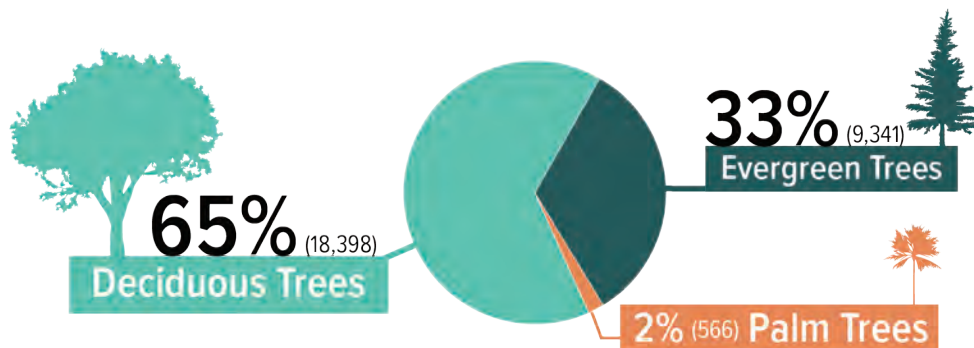


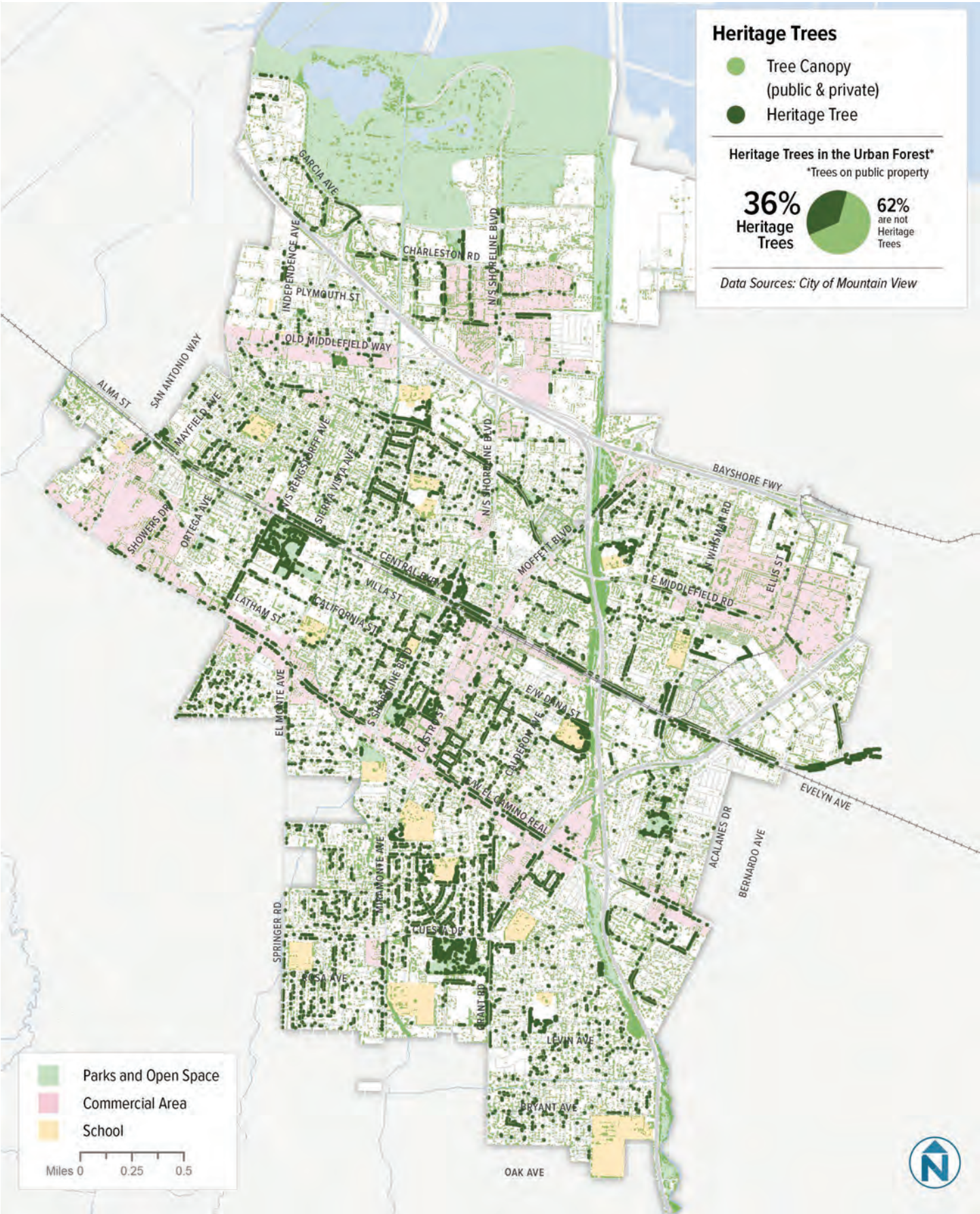
Figure 36 Street Tree Count by Type



Heritage Trees

The city has designated 9,297 community trees as heritage trees, and 7,332 are street trees. A "Heritage Tree" is defined as any tree that has a trunk circumference of 48 inches or more measured at fifty-four inches (54") above grade. Three genera, oak (*Quercus*), redwood (*Sequoia*), and cedar (*Cedrus*), are considered Heritage if they have a circumference of 12 inches or more.

Figure 37 Heritage Trees



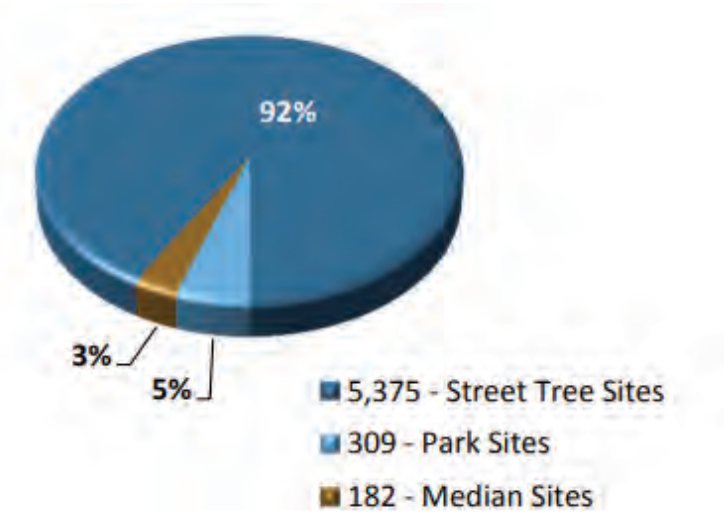
Street Tree Planting and the Urban Forest

The City of Mountain View has made considerable effort in planting and nurturing an urban forest of over 230 tree species.

Like many Cities in California, however, most tree and shrub species planted along City streets are non-native species. All street trees provide important shade, cooling, stormwater, water pollution reduction, and air purification services. However, non-native trees and shrubs like London Plane trees, Chinese Pistache, and Ginkgo do not provide the critical biodiversity and habitat benefits of local native species like Coast Live Oak, Valley Oak, Big Leaf Maple, Arroyo Willow, California Laurel, California Buckeye, and Box Elder.

Street tree planting sites represent a significant opportunity for increasing the City's canopy cover and increasing the proportion of local native species. According to the CTMP, street tree planting sites account for 92 percent of vacant planting sites in the City (see Figure 42). Strengthening the City's street tree presence and increasing the proportion of native species can allow the city to reach its goal of increasing tree coverage and enhancing biodiversity.

Figure 38 Vacant Planting sites by location



Development Review

The City has a role in maintaining or increasing tree planting and canopy coverage along active transportation networks as part of the development entitlement process. Requirements for developers to preserve or plant trees affect the comfort and amenity of pedestrian facilities in Mountain View as well as the potential for transportation networks to contribute to broader environmental goals.

Development projects are reviewed by City staff relative to approved and posted standards and plans such as:

- Mountain View City Code (MVCC) sections such as MVCC Chapter 36 on Zoning, which outlines specifications such as lot coverage, setback and floor-area-ratio (FAR);
- Precise Plans such as the North Bayshore Precise Plan, which outlines requirements related to a Habitat Overlay Zone, and
- 2019 Standard Provisions and Details, which outline details such as sidewalk width, lane widths and visibility triangles at driveways and corners.

Standards and regulations such as those listed above provide clarity for developers and also have implications for achieving goals related to active transportation, green streets and biodiversity. For example, Mountain View design standards for redevelopment of existing sites typically result in a consolidation and reduction in the number of driveways for most land uses especially on large transportation corridors, which can increase the area for green stormwater infrastructure, landscaping and street trees.

Street Tree Planting Types

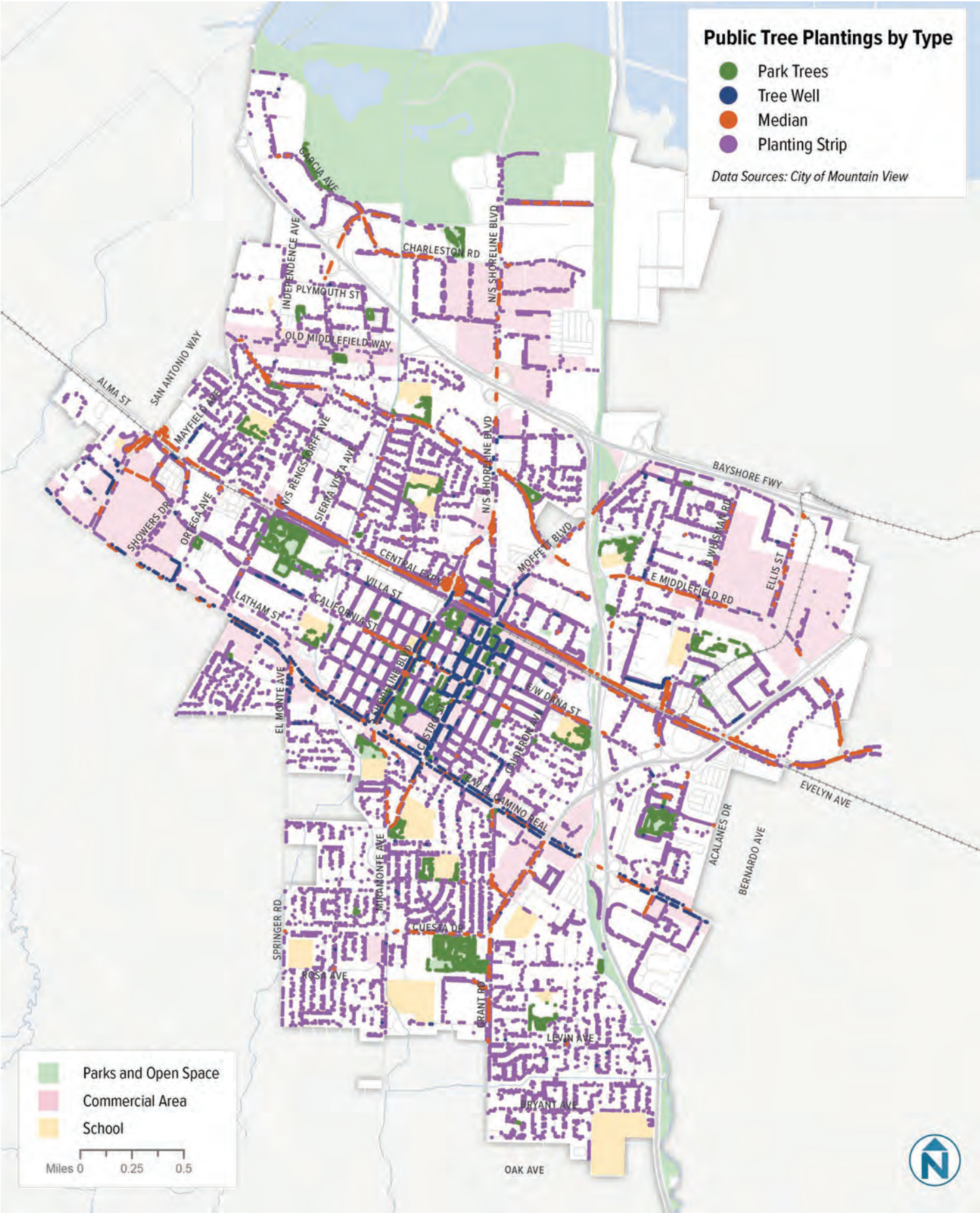
Over the years, the city has successfully planted public trees along streets, including plantings in tree wells, planting strips, medians, parking spaces, and cul-de-sacs, each offering unique challenges and opportunities. Based on the City's Urban Forest GIS Data, existing street tree planting locations in Mountain View are categorized by the following type of planting:

- Planting Strips (67%)
- Parks (16%)
- Medians (12%)
- Tree Wells (5%)

While the city's data categorizes planting locations into four primary groups, this Existing Conditions report delves deeper into trees planted within the public right-of-way, differentiating planting types into eight categories. A brief explanation of each planting type is provided after Figure 40 including the associated challenges and opportunities.

1. Planting Strip - Detached Sidewalks and Curb Extension
2. Planting Strip - Attached Sidewalks
3. Medians
4. Protected Intersections and Bikeways (Class IV)
5. Cul-de-sac
6. Tree Wells - Sidewalk
7. Tree Wells - Parking
8. Phased Greening

Figure 39 Public Tree Planting by Type



1. Planting Strip - Detached Sidewalks and Curb Extensions

Planting strips located between the street and sidewalk along the roadway and at the intersection, provide space for planting vegetation along a road. Depending on their width, they can accommodate plants from towering shade trees to low-lying perennials. These strips physically distance sidewalks from vehicular lanes, narrowing the roadway with trees that create a sense of enclosure, which can help slow traffic and improve safety and comfort for all users.

Challenges: The city maintains designated street trees; all other plant material in the planting strip is the adjacent property owner's responsibility. Overgrown vegetation may impede narrow sidewalks. Plants in curb extensions and within sight triangles should be carefully chosen and placed to maintain visibility.

Opportunities: Detached sidewalks and planted curb extensions foster safe and more comfortable pedestrian environments in residential neighborhoods. They provide shade for pedestrians, create a buffer from traffic, slowing traffic, enhancing road safety and encouraging more walking trips.

Figure 40 Examples of Planting Strips with Detached Sidewalks



Top: Morgan Street, Middle: Middlefield Road, Bottom: Oxnard, California, Source Dan Burden BikePedImages.org

2. Planting Strip - Attached Sidewalks

Planting strips between sidewalks and private property often act as buffers between the street and parking lots, buildings, or other private spaces, offering a protective boundary between public and private spaces.

Challenges: Planting strips with attached sidewalks do not provide separation between the street and the sidewalk as detached sidewalks do. Overgrown vegetation may impede narrow sidewalks.

Opportunities: Planting areas along sidewalks creates a sense of enclosure, making pedestrians feel more comfortable and less exposed. Consideration can be given to providing additional space for greening the street.

Figure 41 Examples of Planting Strips with Attached Sidewalks



Left: Morse Avenue, Center: Central Expressway, Right: San Ramon Avenue

3. Medians

Street medians present opportunities for green street elements like trees and vegetation in the center of the roadway. Some large roads, like Central Expressway and West Middlefield Road, have expansive median plantings.

Challenges: Medians may be difficult to maintain due to their location in the middle of the street. They also do not provide the same sense of enclosure, shade, or roadway benefits as planting strips.

Opportunities: Expanding green street treatments in medians along larger roads can help to reduce vehicle speeds, infuse vibrancy, and contribute to climate and tree canopy goals.

Top: Central Expressway, Bottom: El Camino Real

Figure 42 Examples of Median Plantings



4. Protected Intersections and Bikeways (Class IV)

The space between the bike lane and the motor vehicle travel lane, also called the protective buffer, presents opportunities for green street elements, including trees, vegetation, and bioswales. Protected intersections provide space for low-lying green treatments at the corner island.

Challenges: Protective buffer areas require available roadway space and can increase costs for plant maintenance. Other challenges include plant encroachment upon the bike lane, including tree branches. Low-lying plants should be used in protected intersection corner islands to maintain visibility.

Opportunities: Opportunities include expanding green street treatments to reduce vehicle speeds, infuse vibrancy, contribute to climate and tree canopy goals, and create separation between the bikeway and motor vehicles, thereby increasing comfort and safety for cyclists of all ages and abilities.

Figure 43 Examples of Protected Intersections and Bikeways



Top Left: Charleston Avenue, Top Right: Shoreline Boulevard at Charleston Avenue,
Bottom: Charleston Avenue

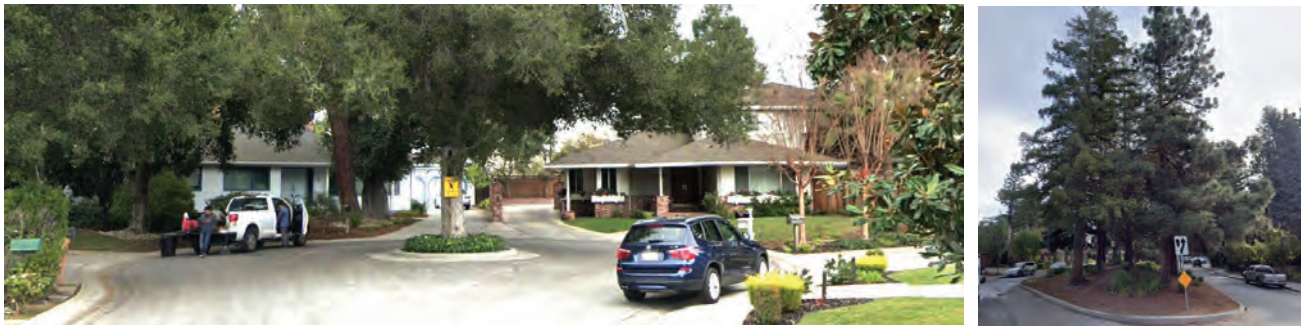
5. Cul-de-sac

Some residential neighborhoods incorporate planting space in the centers of cul-de-sacs and small intersections, introducing greenery and expanding the tree canopy over paved areas. These planting spaces break up expanses of pavement and help maintain slower vehicular speeds. This unique Mountain View example of planting location can serve as a model for other cities.

Challenges: Limited planting area and possibly limited neighborhood support. Difficult establishment period after planting due to limited irrigation.

Opportunities: Transform traditionally non-green cul-de-sacs into lush green spaces. Serve as a model for other cities, with minimal impact on traffic flow.

Figure 44 Examples of Cul-de-Sac Plantings



Left: Damian Way, Right: Hedgerow Court

6. Tree Well – Sidewalk

Tree wells provide planting space within paved areas and are often used in urban settings like downtown areas. Tree wells can be open or covered with tree grates and are typically integrated with sidewalks.

Challenges: Limited planting areas reduce soil volume and room for water infiltration and root growth. Few tree species are well suited for these conditions, so choices can be limited, depending on the size of the planting space.

Opportunities: Plantings in tree wells help increase tree canopy in high-activity areas and create a human-scaled environment and sense of enclosure with trees on both sides of the street.

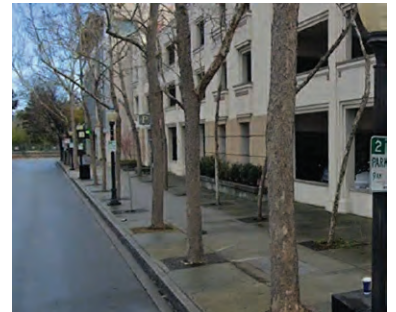
Figure 45 Examples of Tree Wells



Left: Moffett Boulevard



Middle: West Evelyn Avenue



Right: Bryant Street

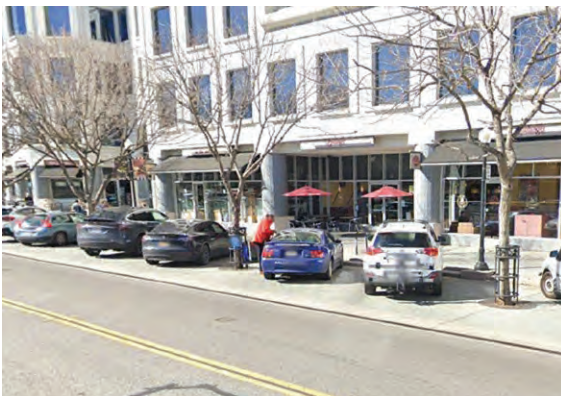
7. Tree Well - Parking Space

Tree wells can also be situated within parking spaces along streets.

Challenges: Tree wells in parking areas may reduce the number of parking spaces and increase the difficulty of changing the roadway layout later.

Opportunities: Tree wells in parking areas provide an innovative way to boost tree canopy on streets without impacting sidewalks. This prioritizes walking space and aligns with climate goals by promoting mode shift and reducing car dependency.

Figure 46 Examples of Tree Wells in a Parking Space



Left: Castro Street, Right: Castro Street



8. Phased Greening

Temporary planters and parklets have gained prominence as strategies to increase greenery and placemaking along streets, particularly in response to the COVID-19 pandemic when cities opened streets for people and closed them for cars. These serve as valuable tools for piloting ideas and can offer interim planting spaces.

Challenges: Existing infrastructure may pose an obstacle to temporary planters and parklets, and parking may be limited.

Opportunities: These strategies provide short-term opportunities to enhance the vibrancy and comfort of the street. Strategic placement can leverage existing planting spaces.

Figure 47 Examples of Phased Greening



Left: Castro Street (east side),



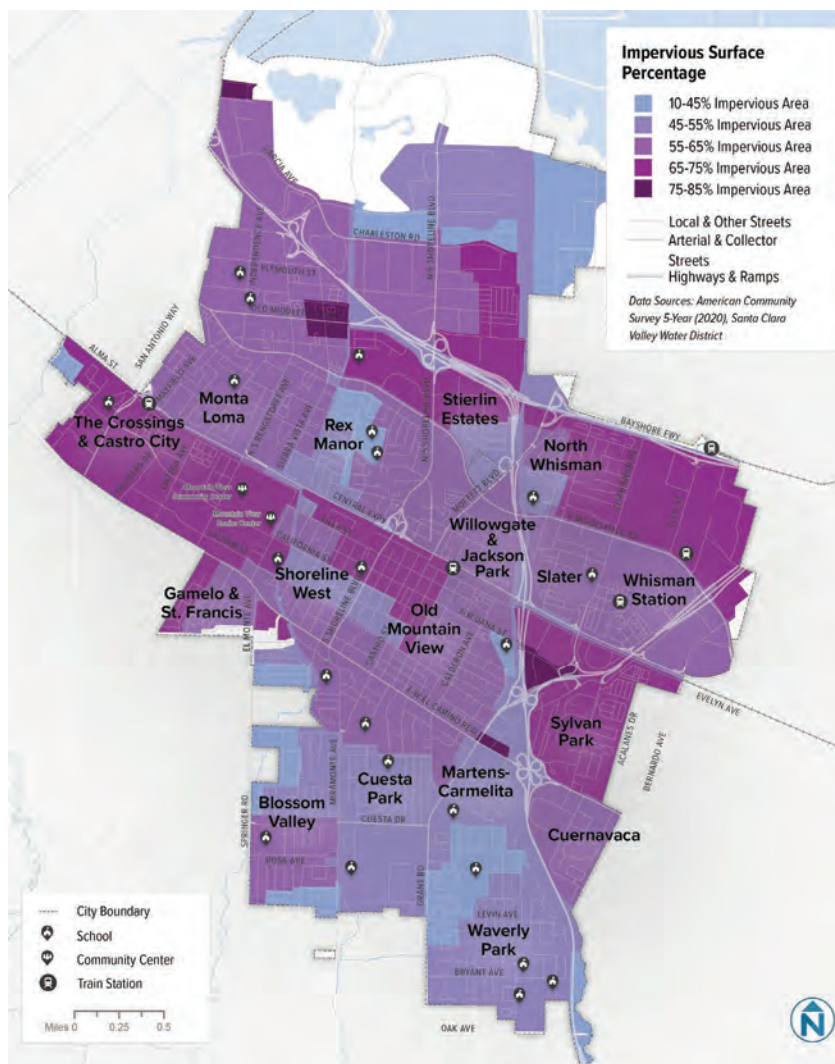
Right: Castro Street (west side)

Impervious Surface Cover

Impervious surfaces include concrete and asphalt surfaces, roofs, and other hard materials that do not allow water to soak into the ground. In contrast, porous surfaces, like gardens and other unpaved areas, allow water to soak into the ground. As cities grow, they tend to have more impervious surfaces in the form of roads, parking lots, and buildings. Impervious surfaces are associated with more stormwater runoff, flood risk and water pollution, and higher surface temperatures. Trees and other vegetation can make a positive difference in areas filled with impervious surfaces by soaking up rainwater and cooling down the surroundings.

In Mountain View, different neighborhoods have varying percentages of impervious surfaces, as seen in the Figure 48. More impervious surfaces mean higher surface temperatures and more stormwater runoff.

Figure 48 Percentage of Impervious Surfaces (Map and Table)



Neighborhood	Average Impervious Surface Percentage
Rex Manor	46%
Waverly Park	50%
Blossom Valley	51%
Martens-Carmelita	52%
Cuesta Park	53%
Old Mountain View	55%
North Whisman	58%
Willowgate & Jackson Park	58%
Stierlin Estates	60%
Cuernavaca	60%
Shoreline West	61%
Slater	63%
Whisman Station	63%
Monta Loma	65%
Sylvan Park	67%
Gamelo & St. Francis	67%
Castro City & The Crossings	68%

6. CODE REVIEW

As part of the MVATP, a review was conducted of existing City documents and codes that set policy and criteria that determine the final form of the built environment. The Code Review included:

- 2002 Standard Design Criteria (SDC)
- [2019 Standard Provisions and Standard Details](#) (SP&D)
- [Mountain View City Code](#)

The findings from this code review can be used as a starting point for updating the City's standards. Mountain View's FY2627 Recommended Budget includes Project 27-27, Mountain View Standards Update. The project will update current standard details for street widths (including lane widths), sidewalks and landscape strip widths, bike lane widths, and drop off duckouts and GSI in landscape strips.

the City may consider updating the following codes and standards to better support bicyclists and pedestrians:

Table 9: Code Review Summary

2002 Standard Design Criteria	
Item	Notes
Chapter 7 - Street Lighting	
Section 3.5 Curb, Gutters, Sidewalk and Driveways • Section 3.5.1 - commercial driveways • Section 3.5.2 curb radius	Commercial driveway maximum width 35 ft Curb return minimum radius 30 ft
Chapter 7 - Street Lighting	
2019 Standard Provisions and Standard Details	
Item	Notes
Chapter 36 Street Trees	
Standard Detail E-11 Traffic Signal Detector Loops	Provides details for in-ground detector loops
Standard Details A-1 through A-9 Sidewalk and street cross-sections	Define typical street cross-sections and driveway details

Curb Ramp Details	Mountain View refers to Caltrans Standards
Standard Details A-22 Side Street/Driveway Pedestrian and Vehicle Triangle of Safety	Defines vehicular and pedestrian triangle of safety for high volume, low volume, residential, and commercial driveways
City Code	
Item	Notes
Section 36.22.50 Required Number of Parking Spaces, Bicycle Parking	Provision of bike parking is a function of vehicle parking
Section 36.32.85 Bicycle parking facilities	Details bicycle parking design standards